

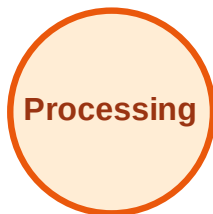
BFS Traversal

Step 1: Start at A

Legend



Complete



Processing



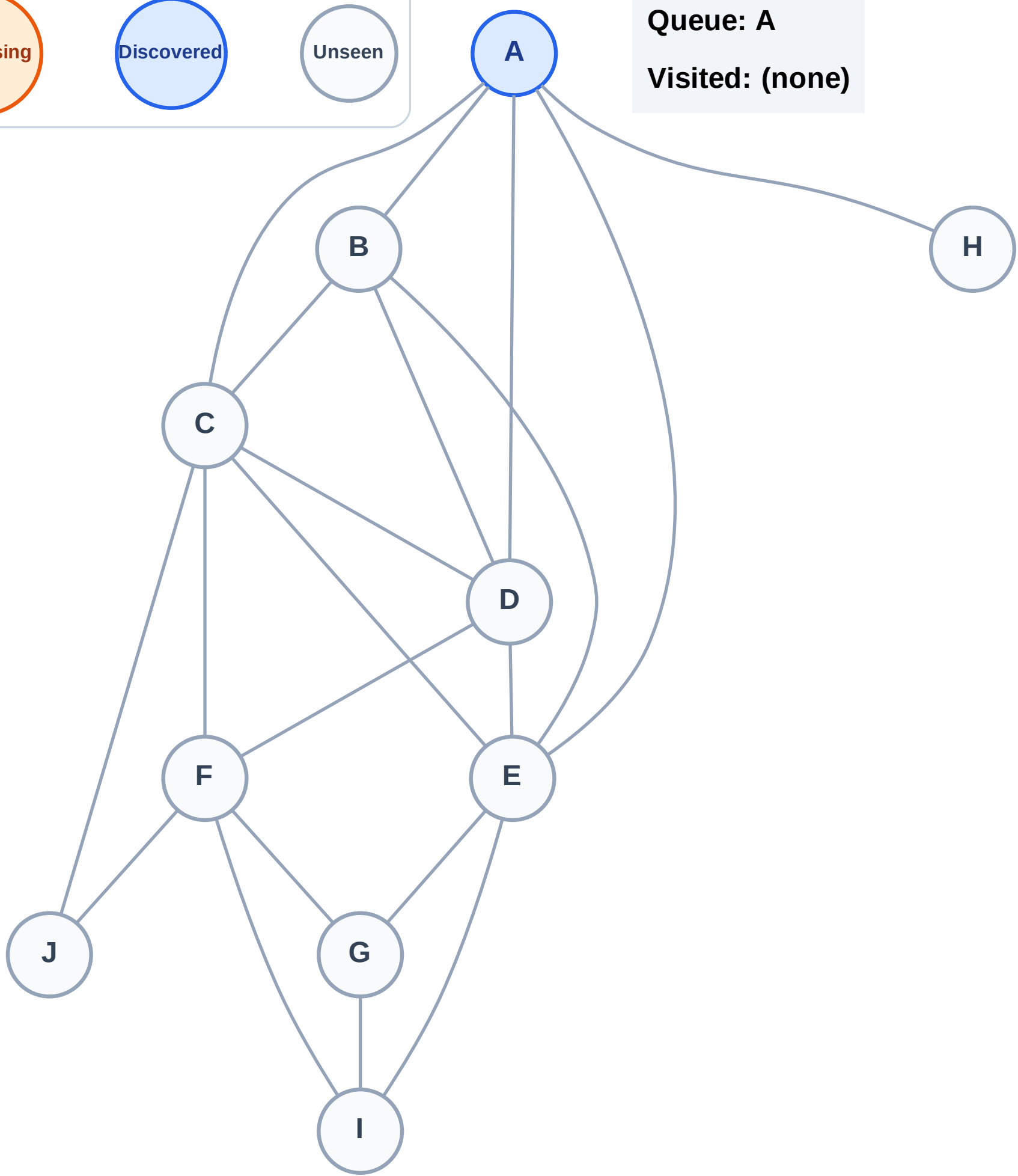
Discovered



Unseen

Queue: A

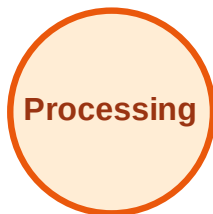
Visited: (none)



BFS Traversal

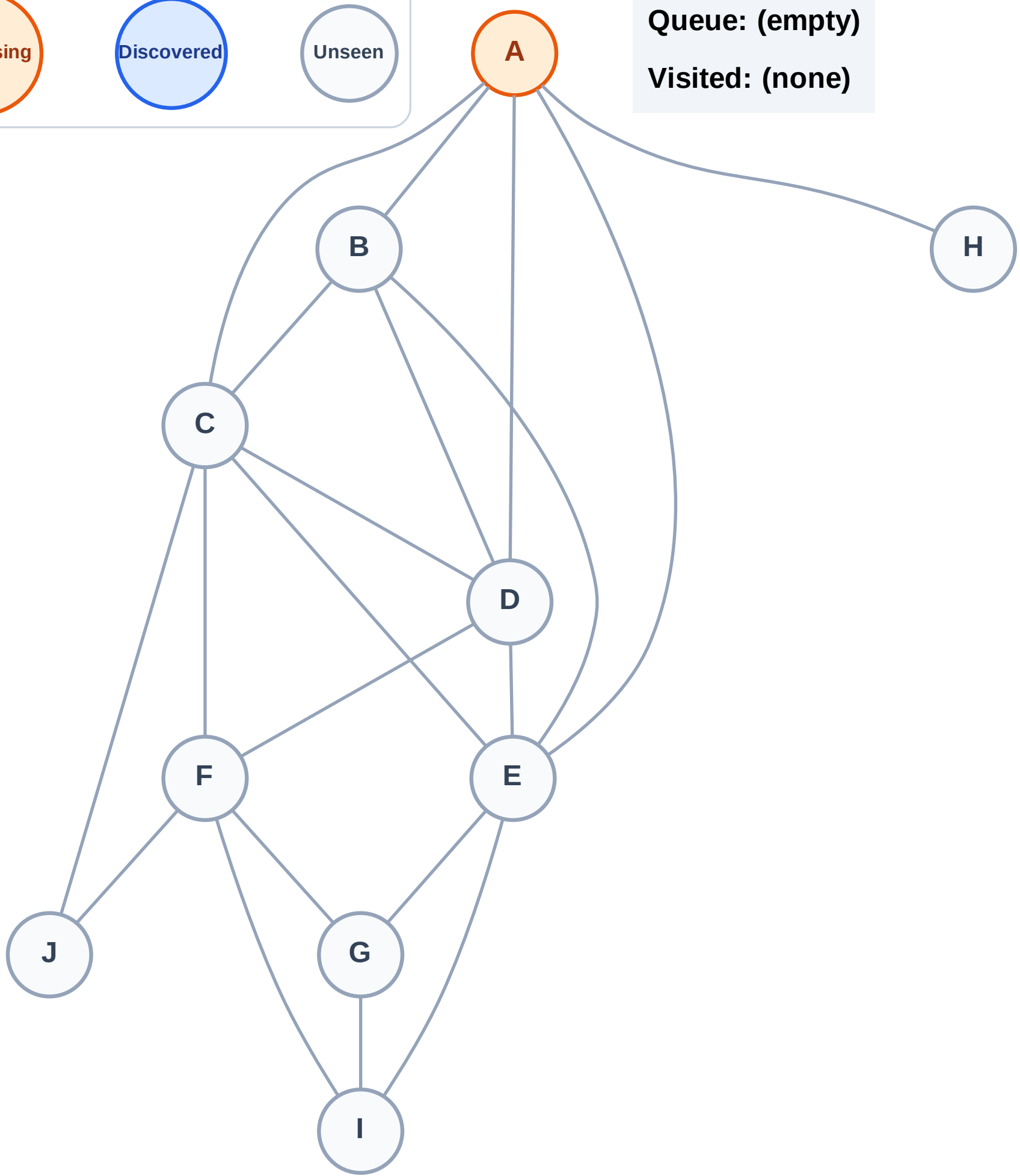
Step 2: Process node A

Legend



Queue: (empty)

Visited: (none)



BFS Traversal

Step 3: Discover B, C, D, E, H from A

Legend

Complete

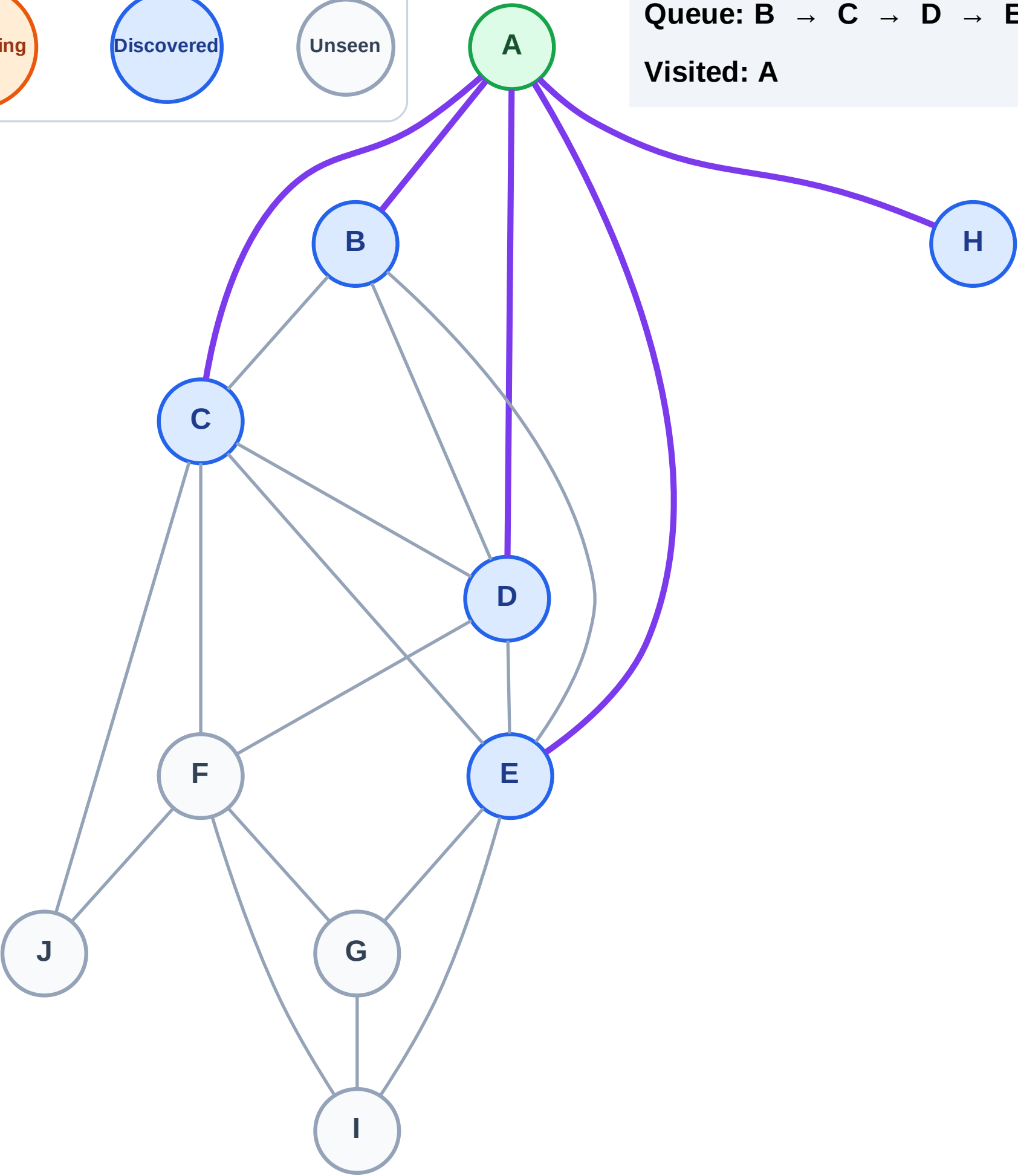
Processing

Discovered

Unseen

Queue: B → C → D → E → H

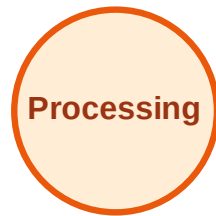
Visited: A



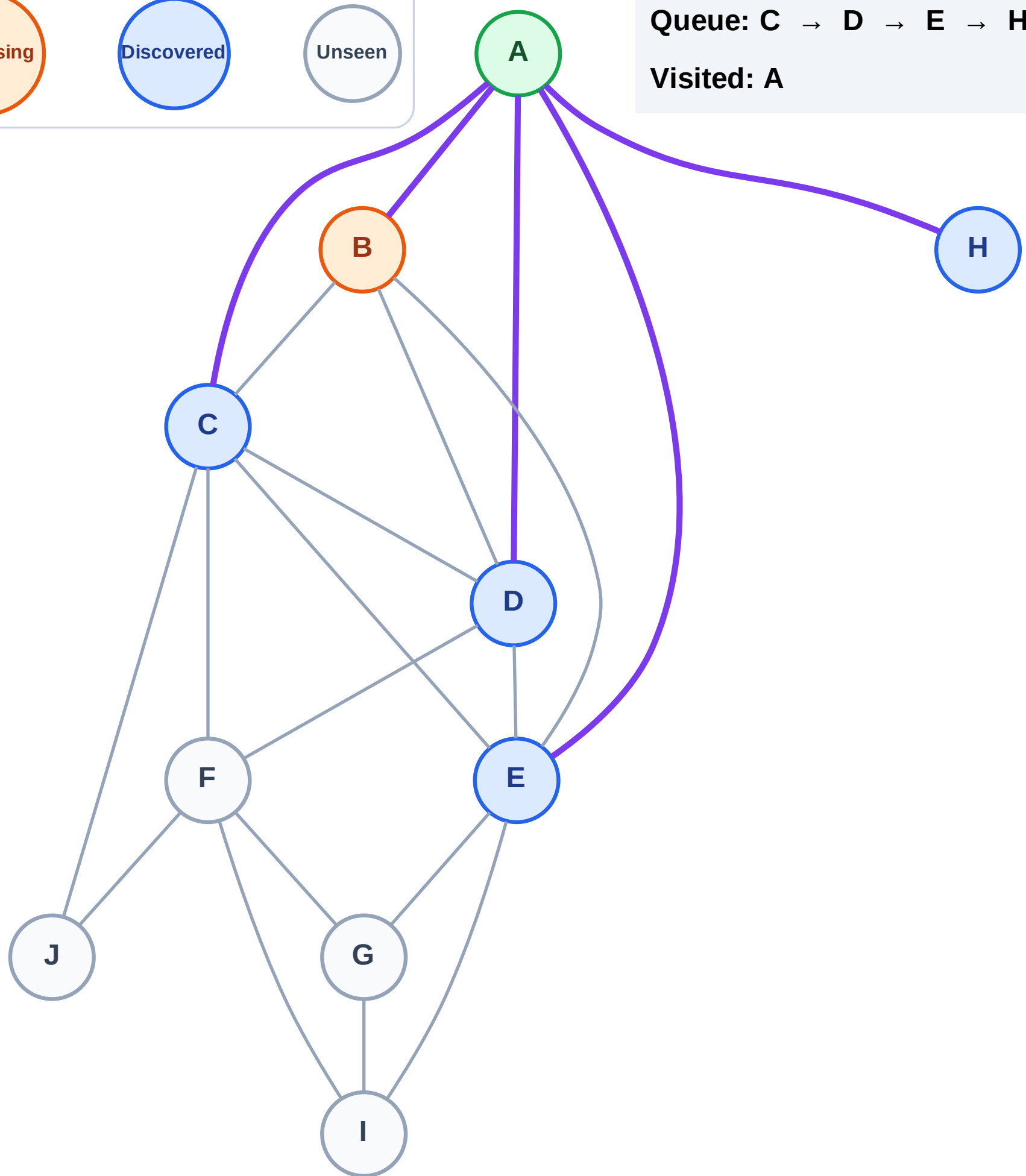
BFS Traversal

Step 4: Process node B

Legend



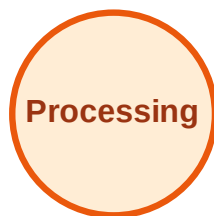
Queue: C → D → E → H
Visited: A



BFS Traversal

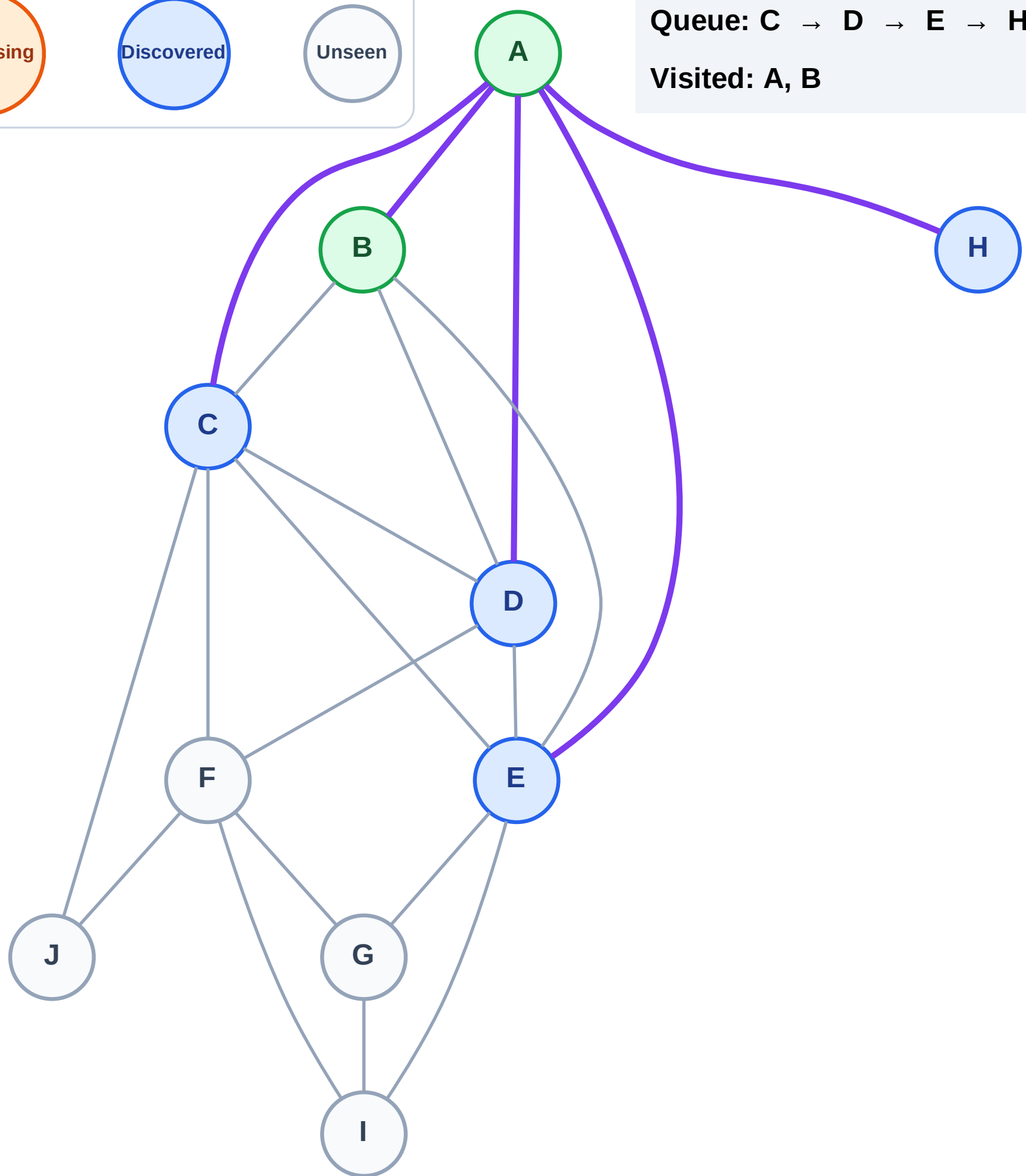
Step 5: No new neighbors from B

Legend



Queue: C → D → E → H

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

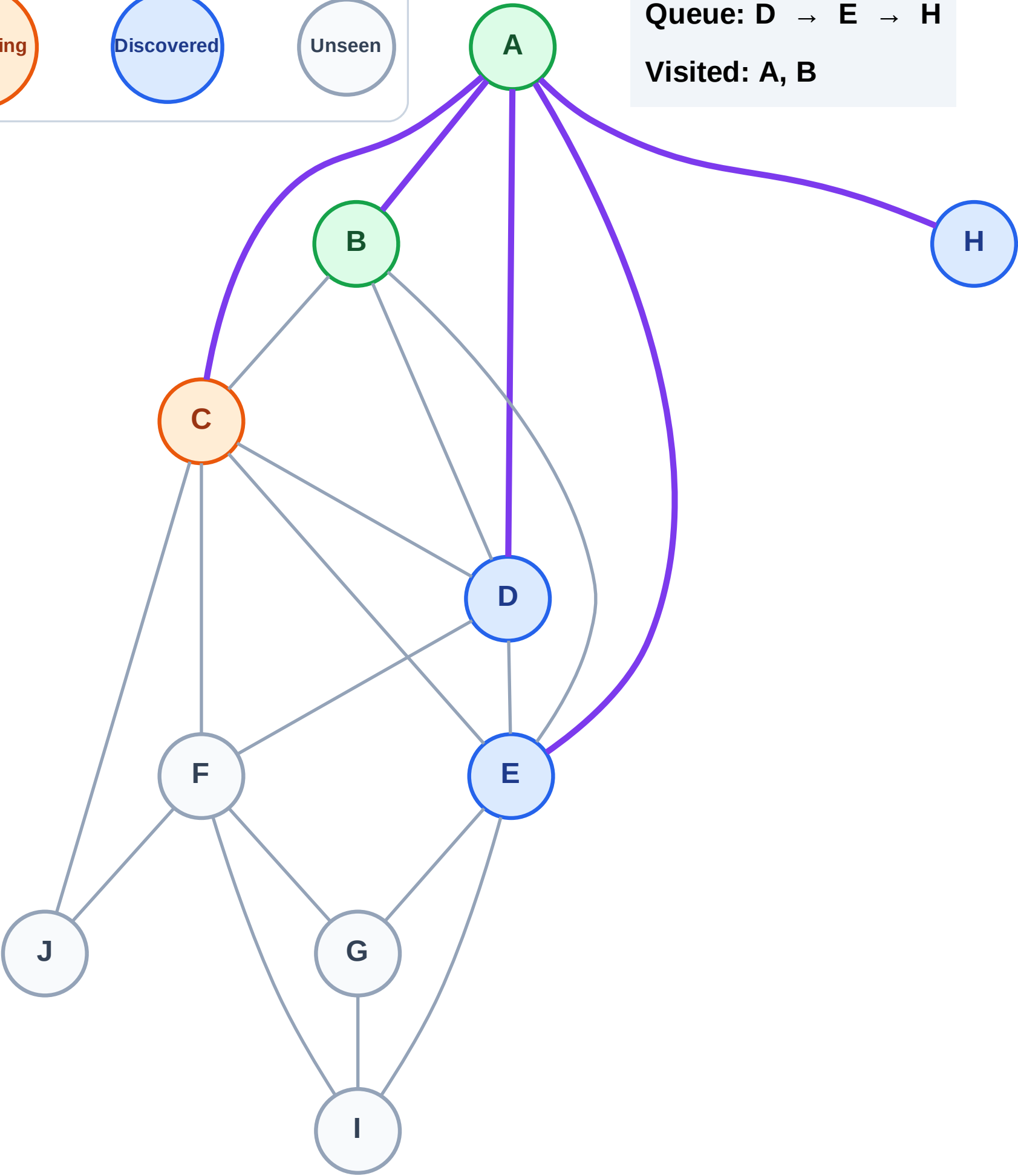
Processing

Discovered

Unseen

Queue: D → E → H

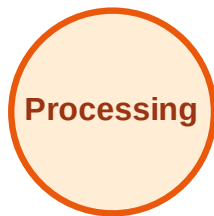
Visited: A, B



BFS Traversal

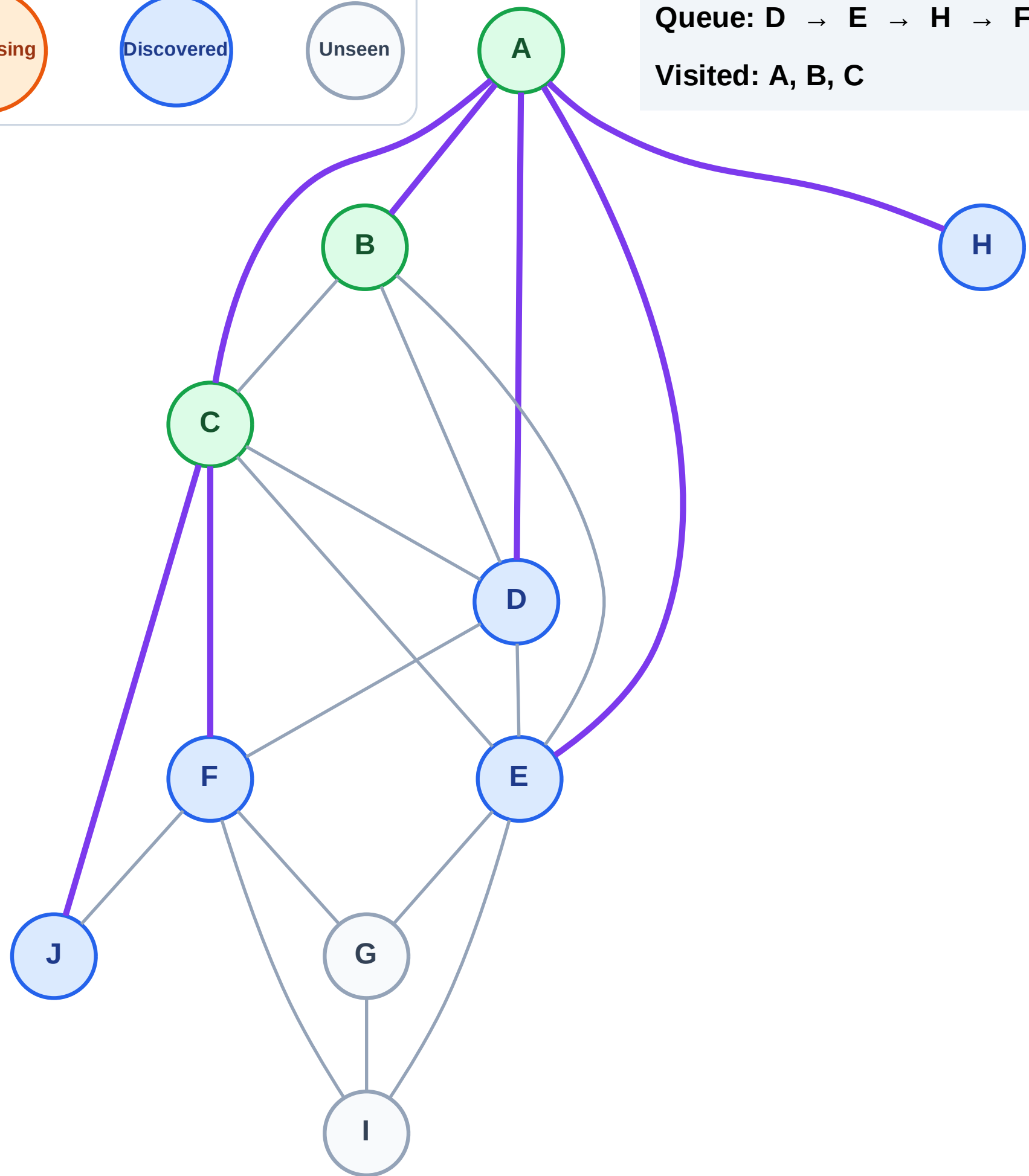
Step 7: Discover F, J from C

Legend



Queue: D → E → H → F → J

Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

Complete

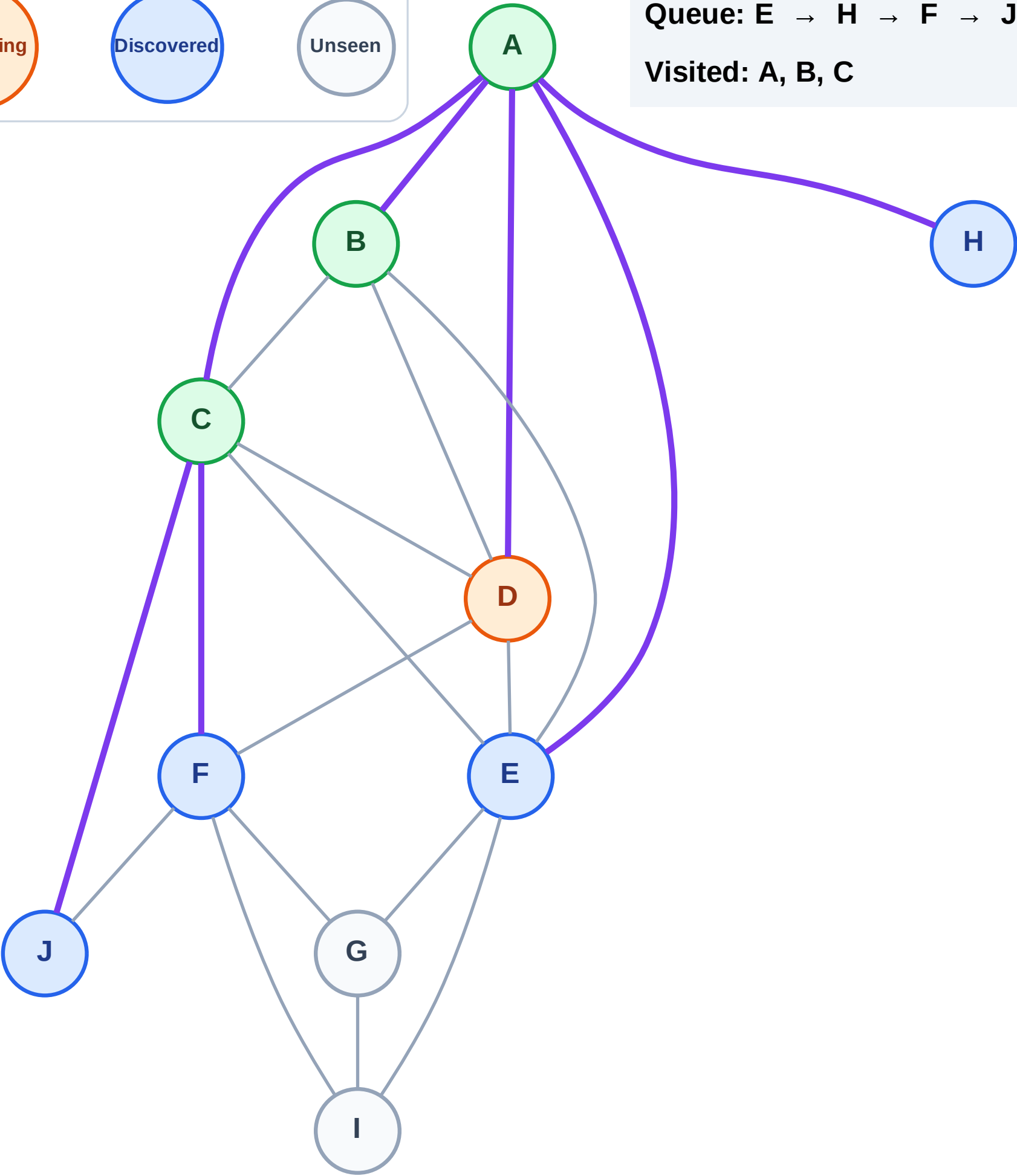
Processing

Discovered

Unseen

Queue: E → H → F → J

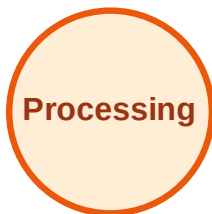
Visited: A, B, C



BFS Traversal

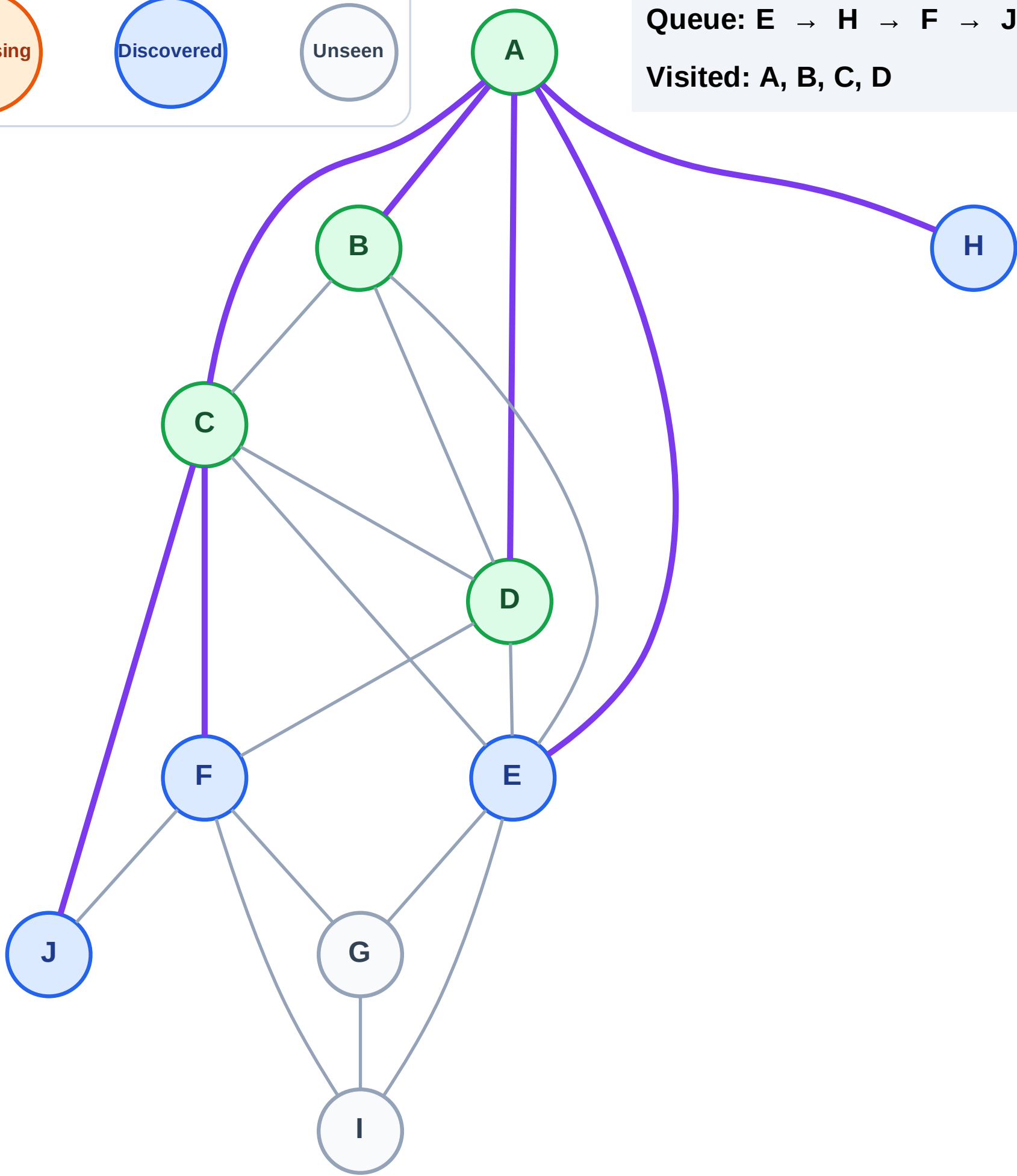
Step 9: No new neighbors from D

Legend



Queue: E → H → F → J

Visited: A, B, C, D



BFS Traversal

Step 10: Process node E

Legend

Complete

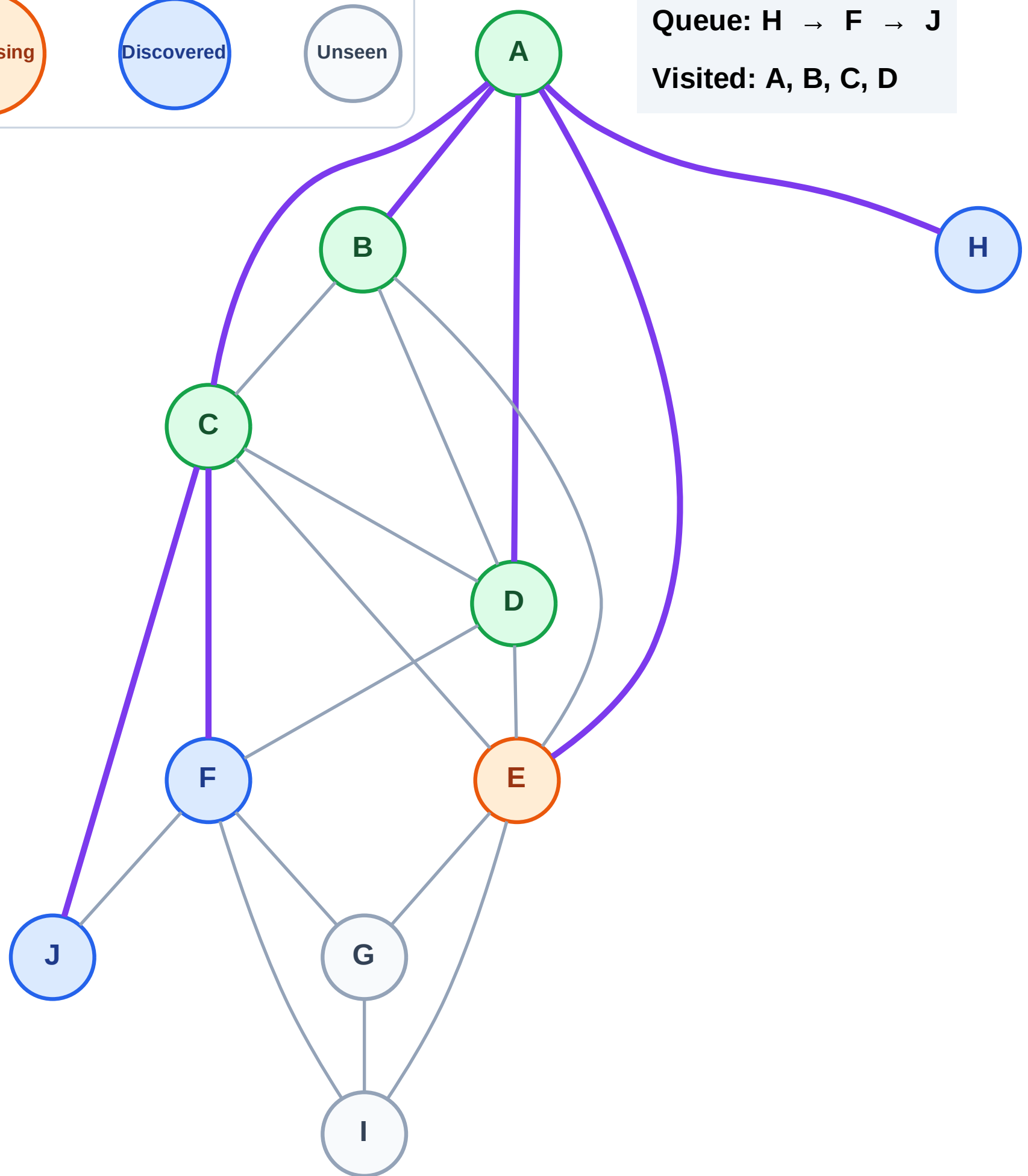
Processing

Discovered

Unseen

Queue: H → F → J

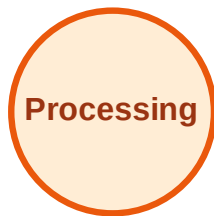
Visited: A, B, C, D



BFS Traversal

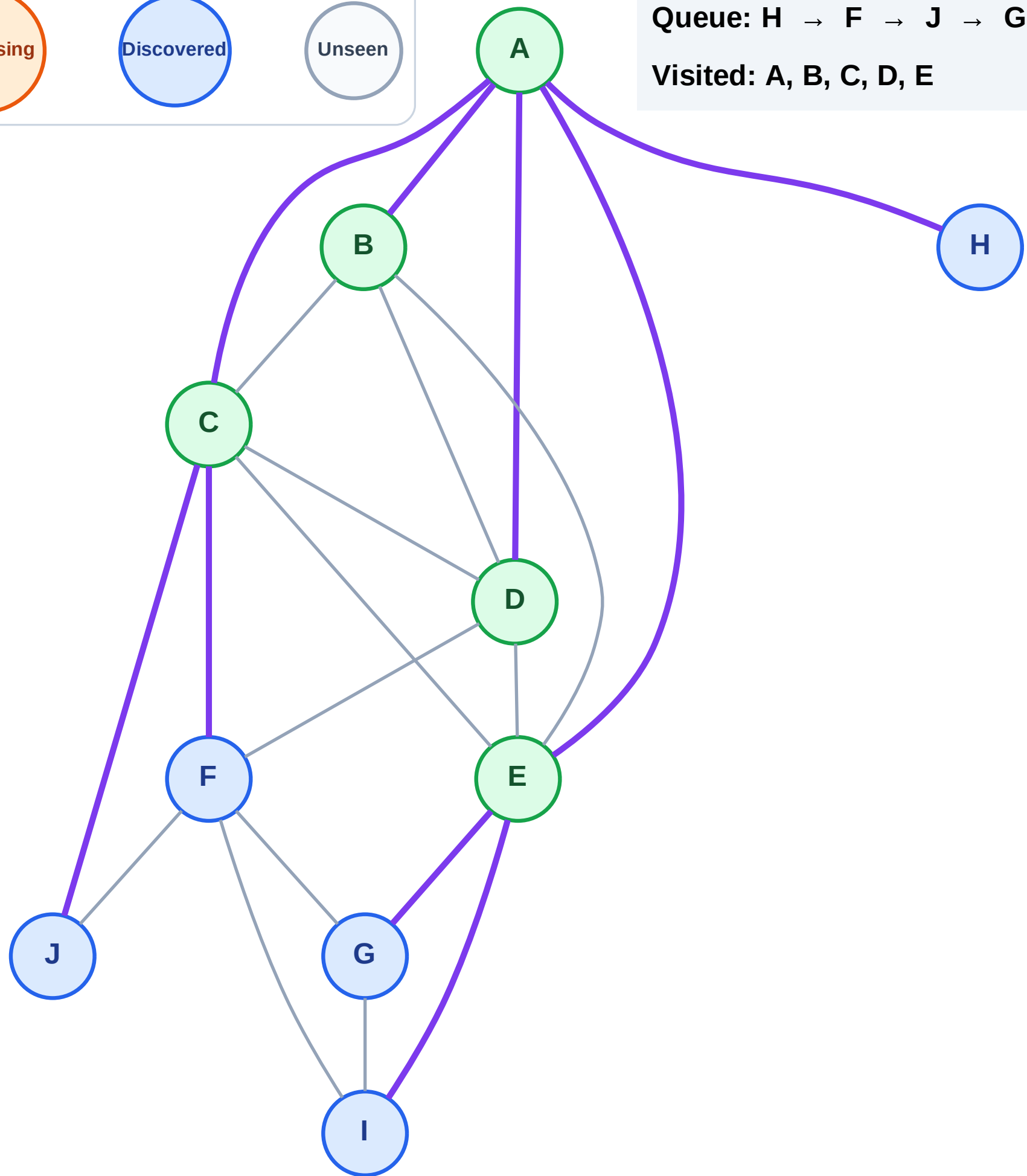
Step 11: Discover G, I from E

Legend



Queue: H → F → J → G → I

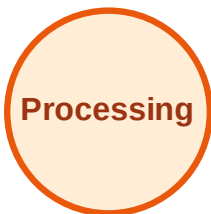
Visited: A, B, C, D, E



BFS Traversal

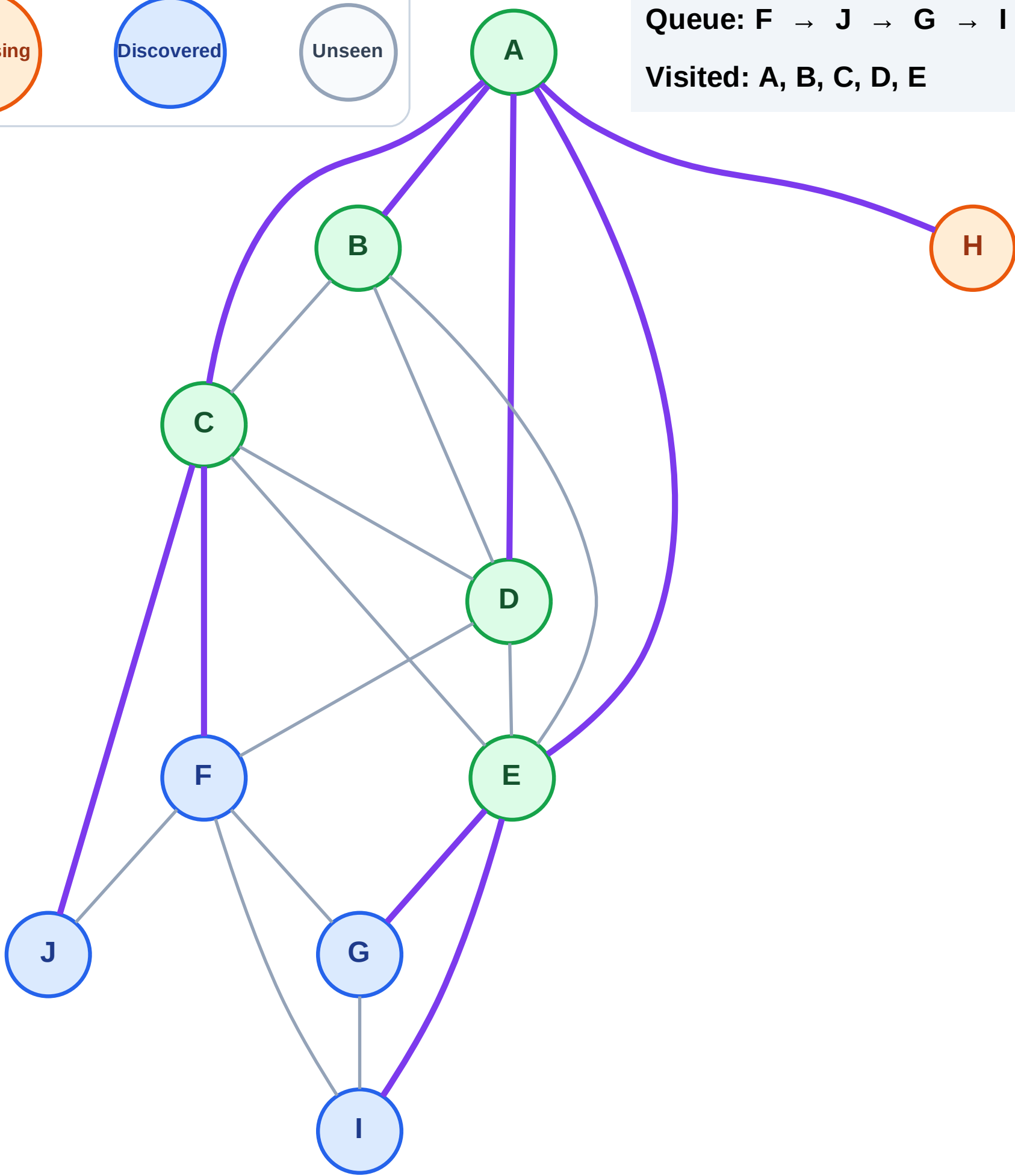
Step 12: Process node H

Legend



Queue: F → J → G → I

Visited: A, B, C, D, E



BFS Traversal

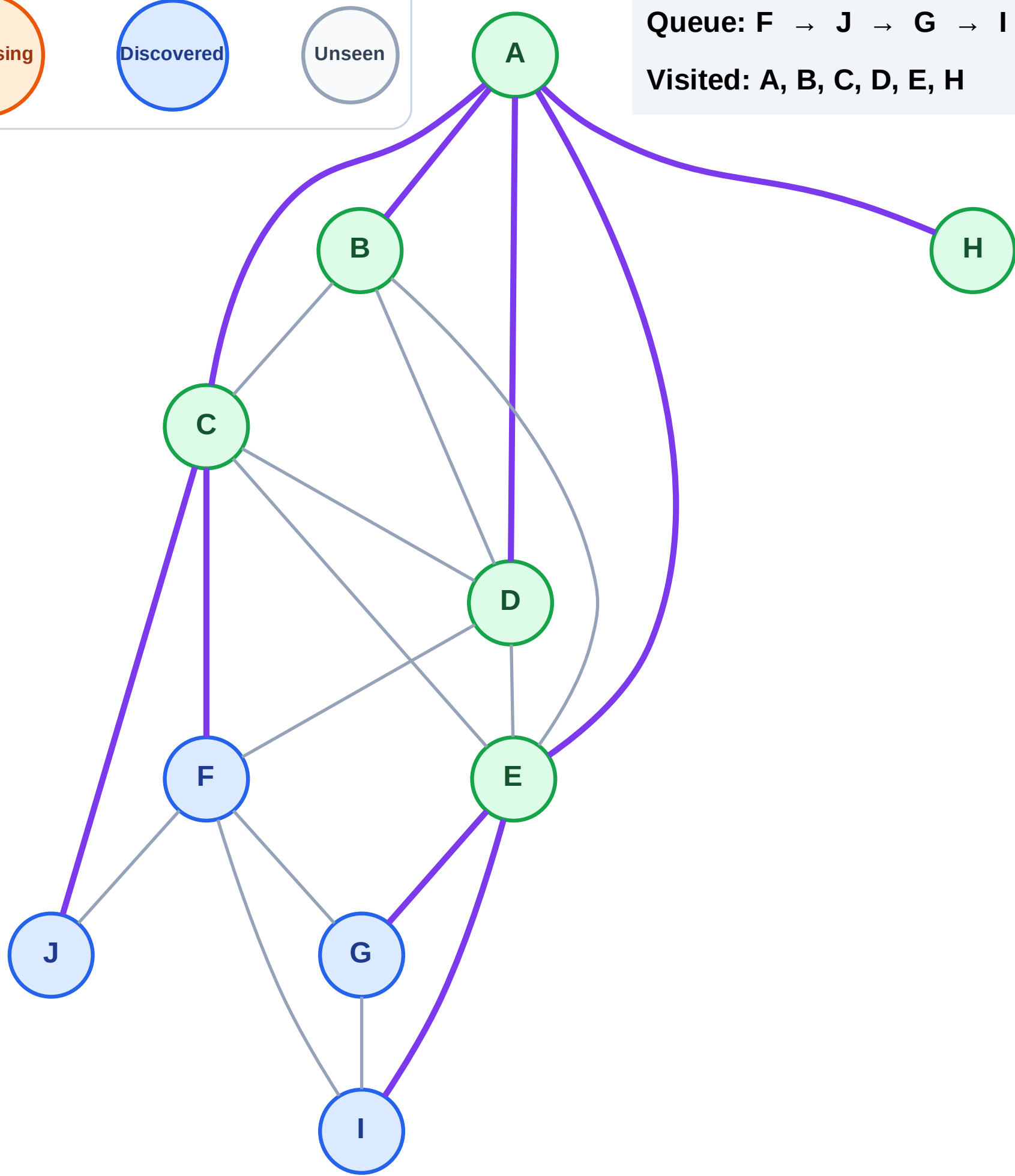
Step 13: No new neighbors from H

Legend



Queue: F → J → G → I

Visited: A, B, C, D, E, H



BFS Traversal

Step 14: Process node F

Legend

Complete

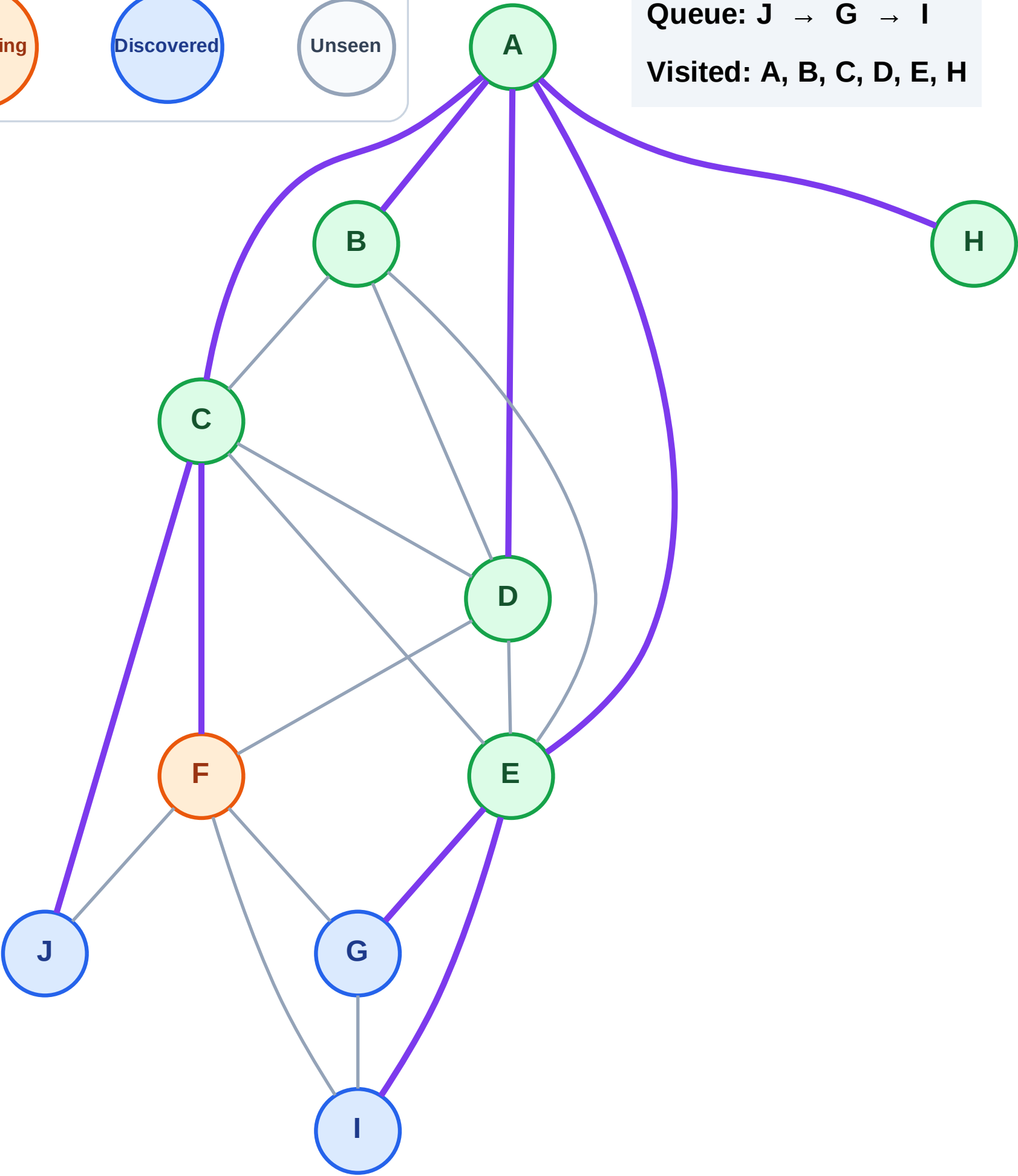
Processing

Discovered

Unseen

Queue: J → G → I

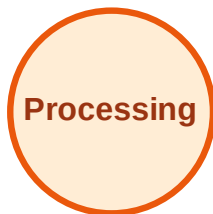
Visited: A, B, C, D, E, H



BFS Traversal

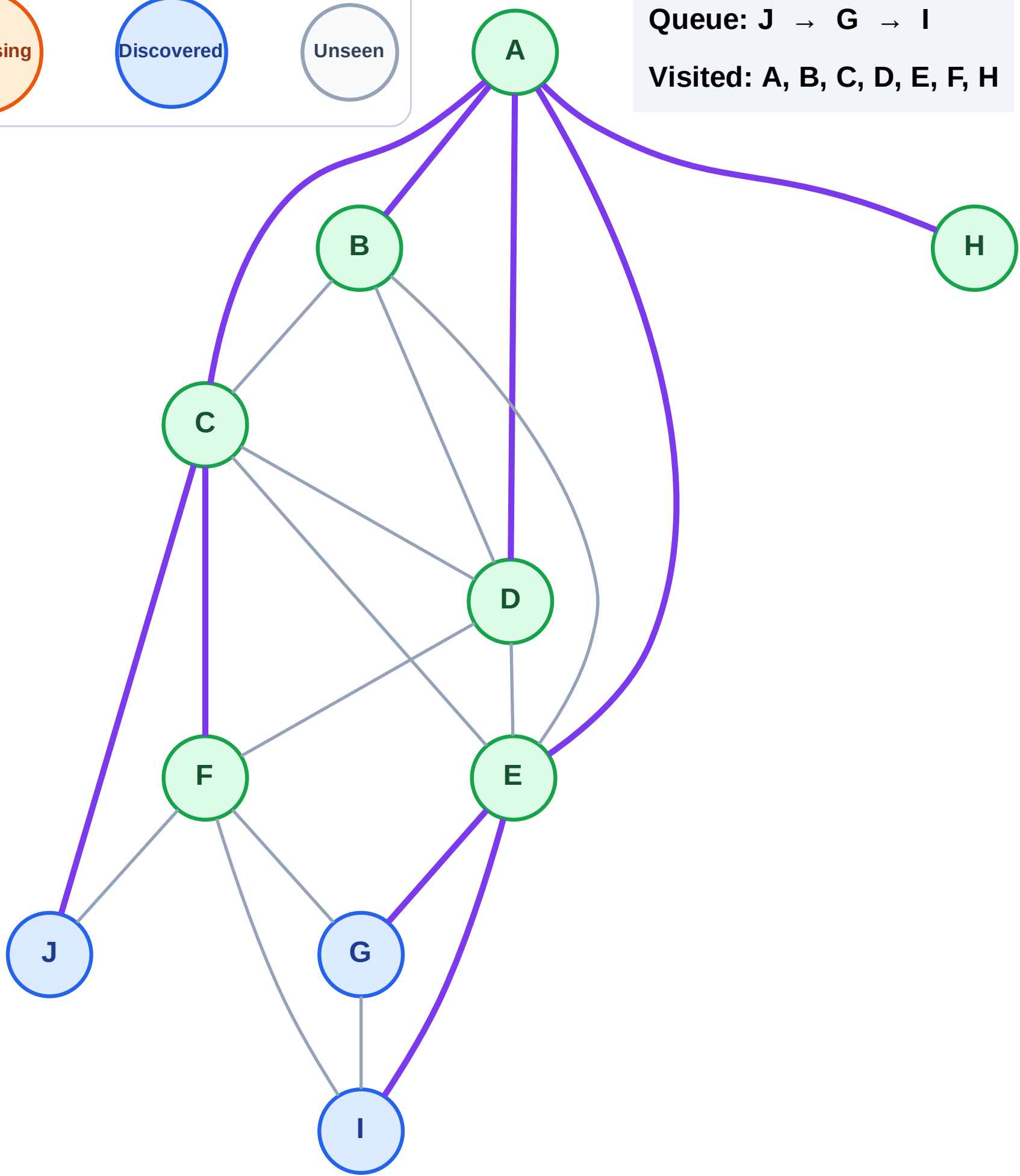
Step 15: No new neighbors from F

Legend



Queue: J → G → I

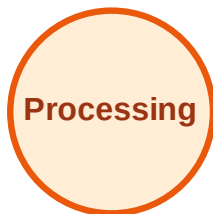
Visited: A, B, C, D, E, F, H



BFS Traversal

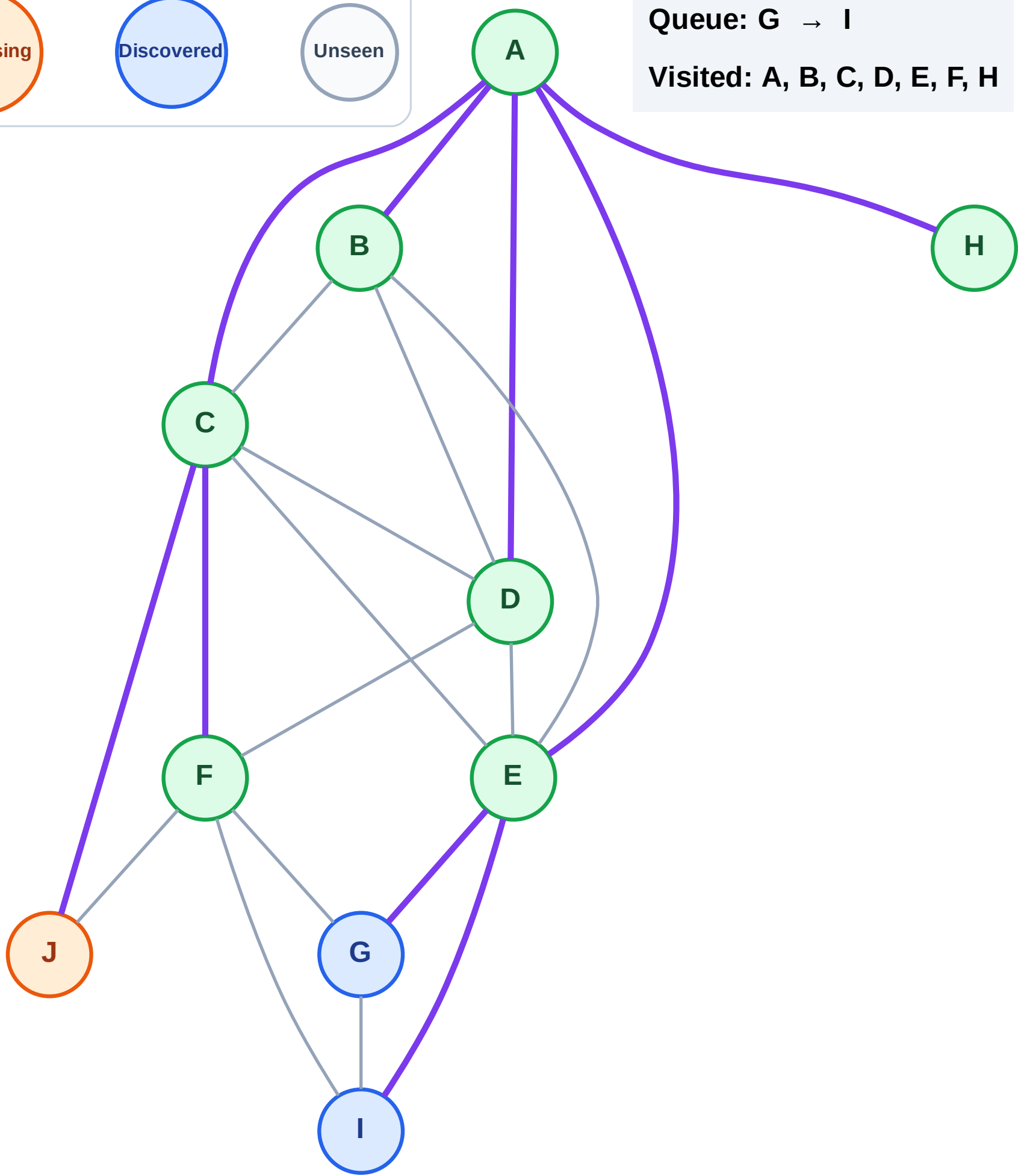
Step 16: Process node J

Legend



Queue: G → I

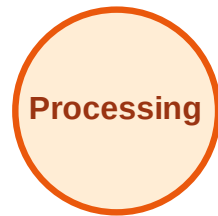
Visited: A, B, C, D, E, F, H



BFS Traversal

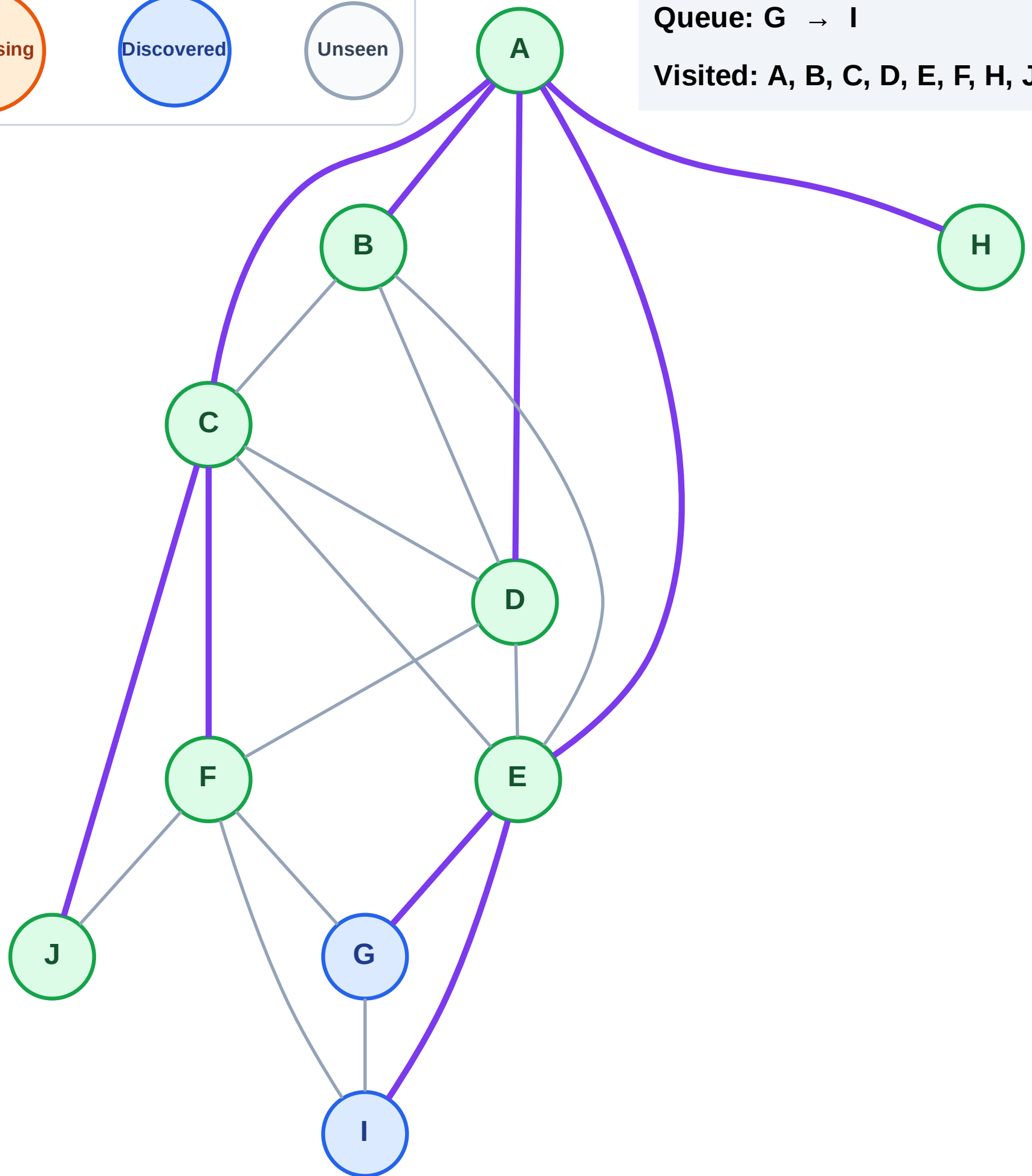
Step 17: No new neighbors from J

Legend



Queue: G → I

Visited: A, B, C, D, E, F, H, J



BFS Traversal

Step 18: Process node G

Legend

Complete

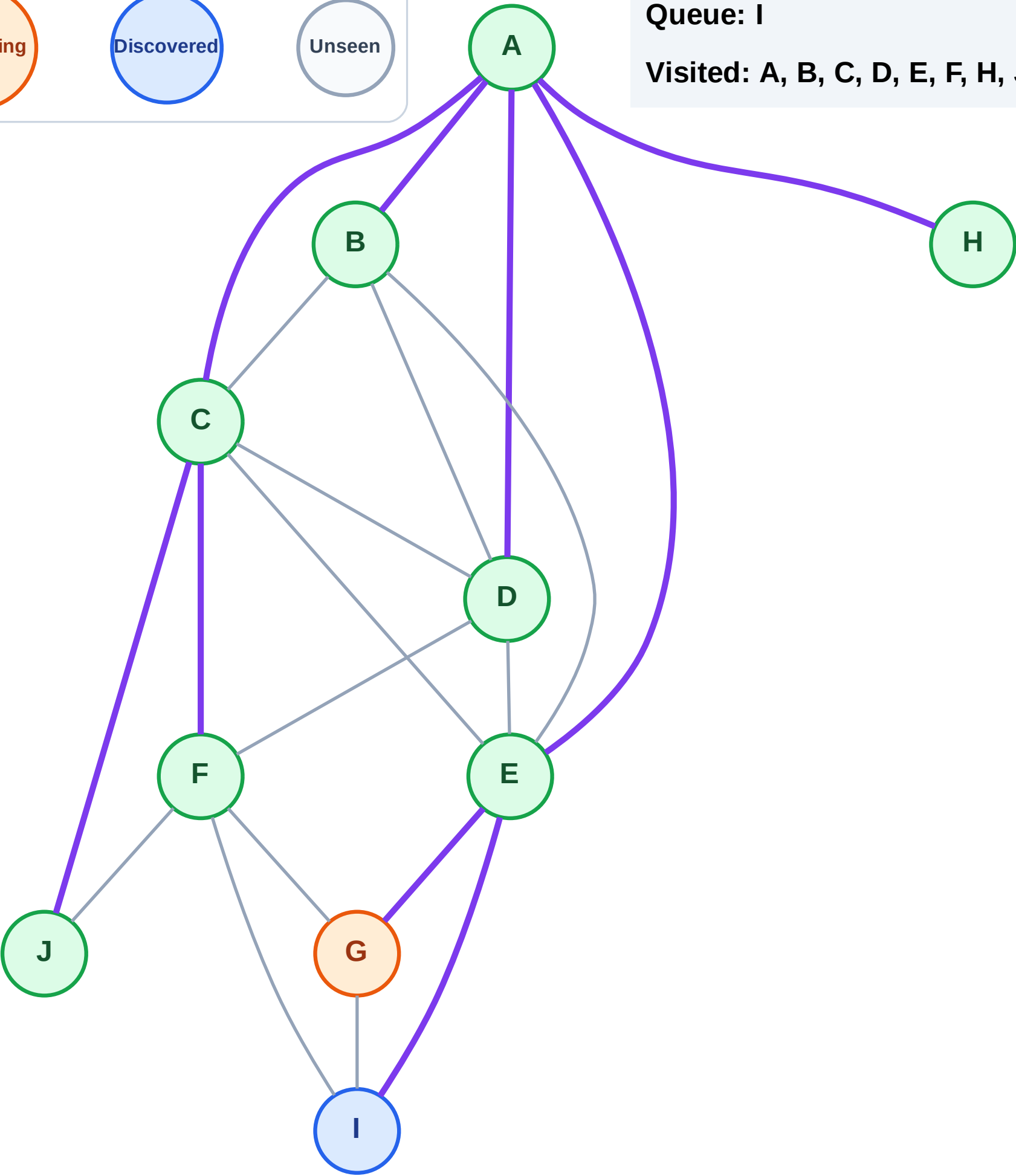
Processing

Discovered

Unseen

Queue: I

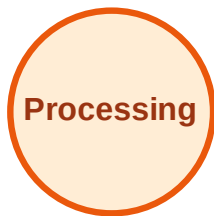
Visited: A, B, C, D, E, F, H, J



BFS Traversal

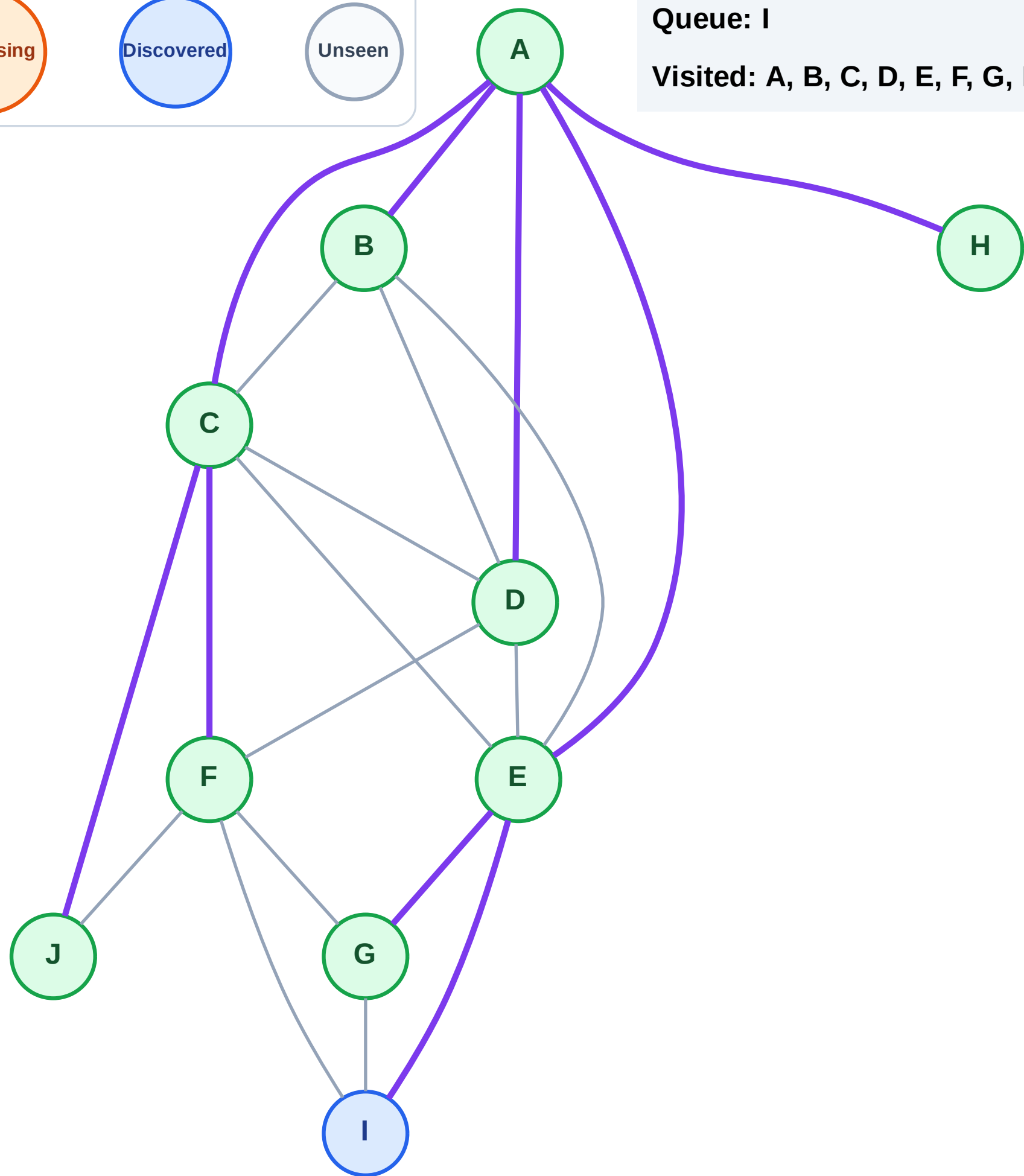
Step 19: No new neighbors from G

Legend



Queue: I

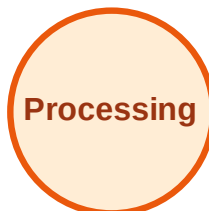
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

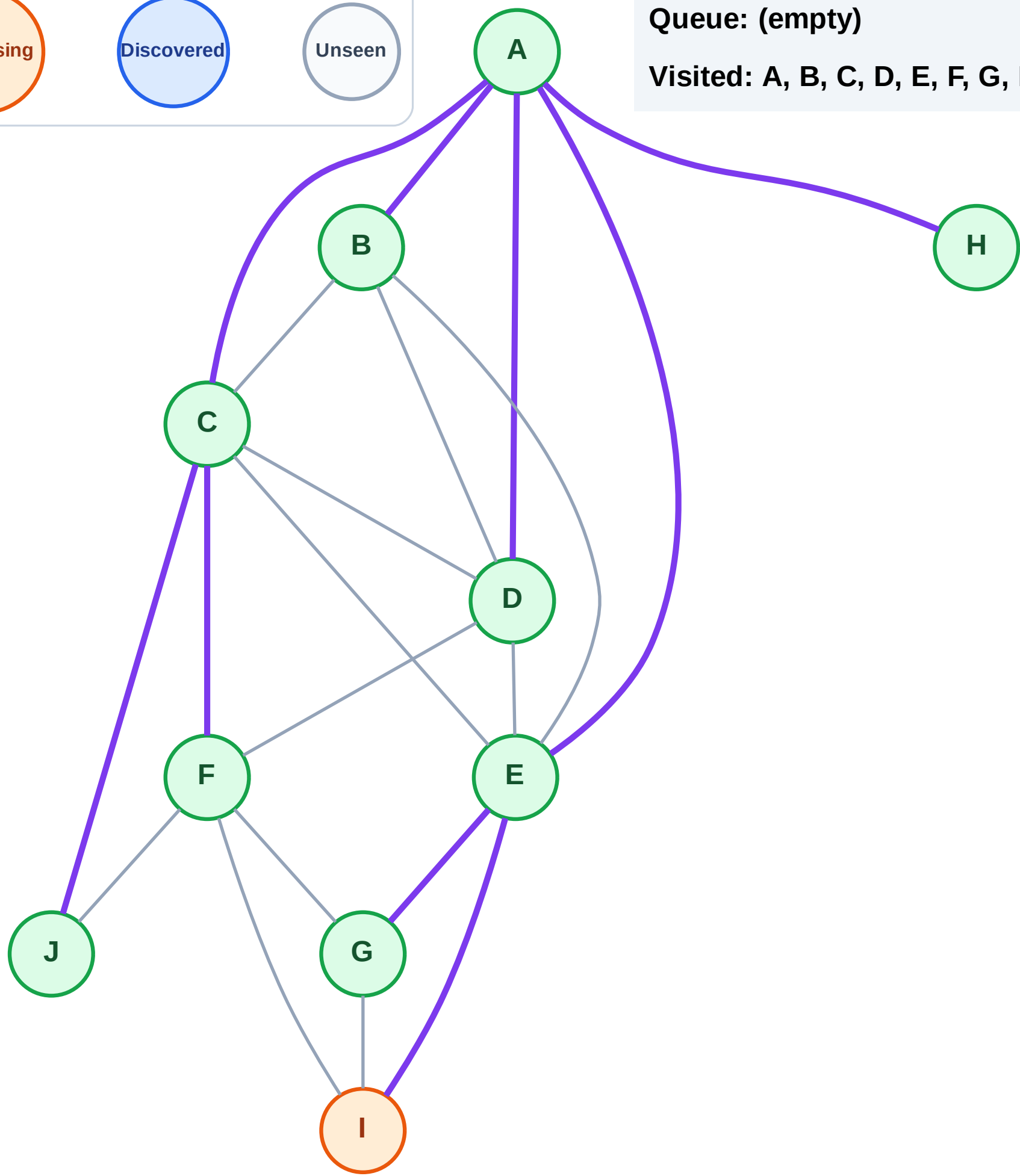
Step 20: Process node I

Legend



Queue: (empty)

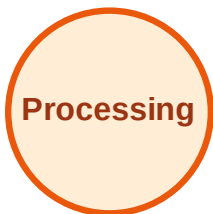
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

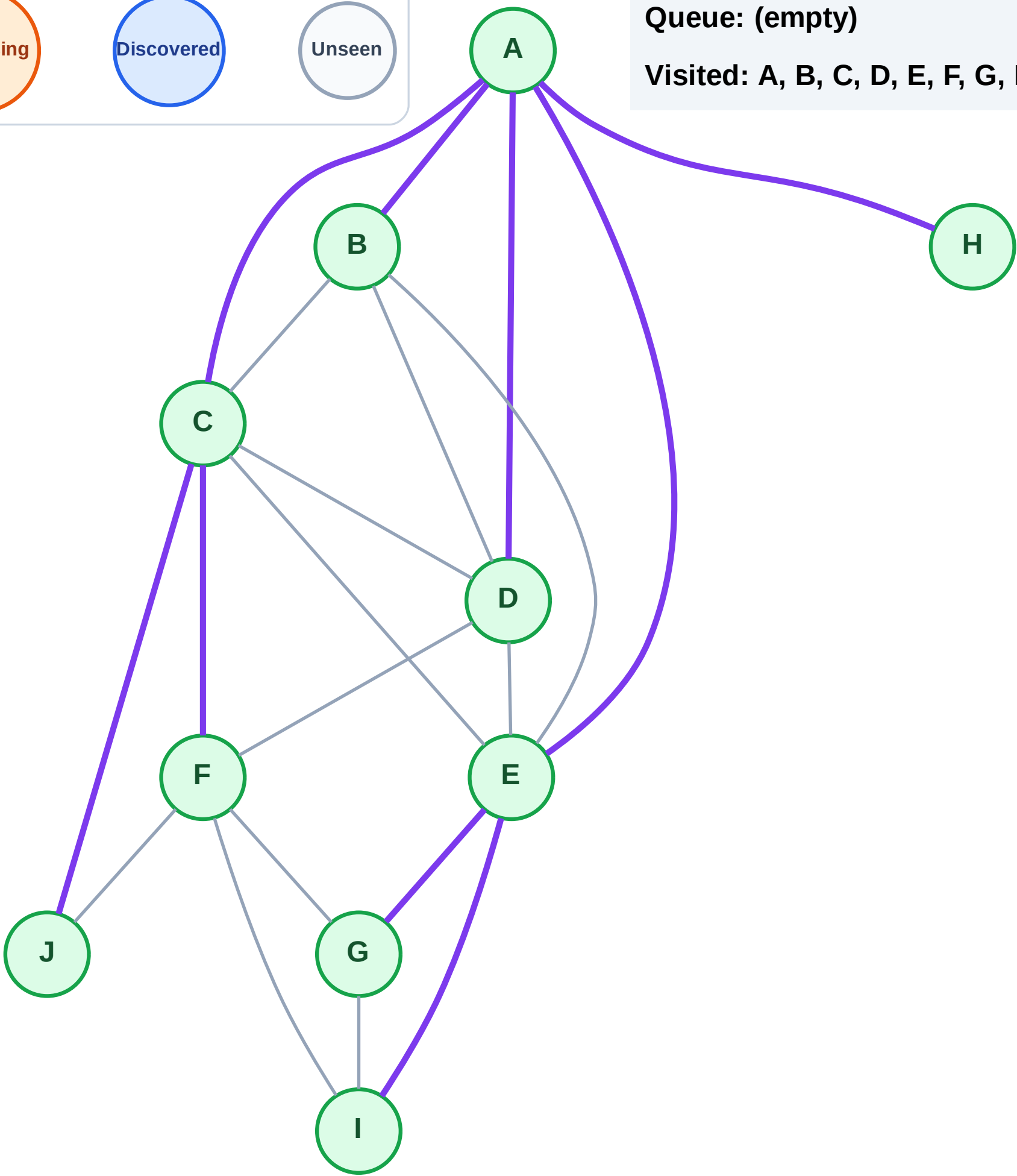
Step 21: No new neighbors from I

Legend



Queue: (empty)

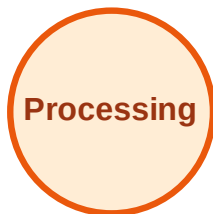
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

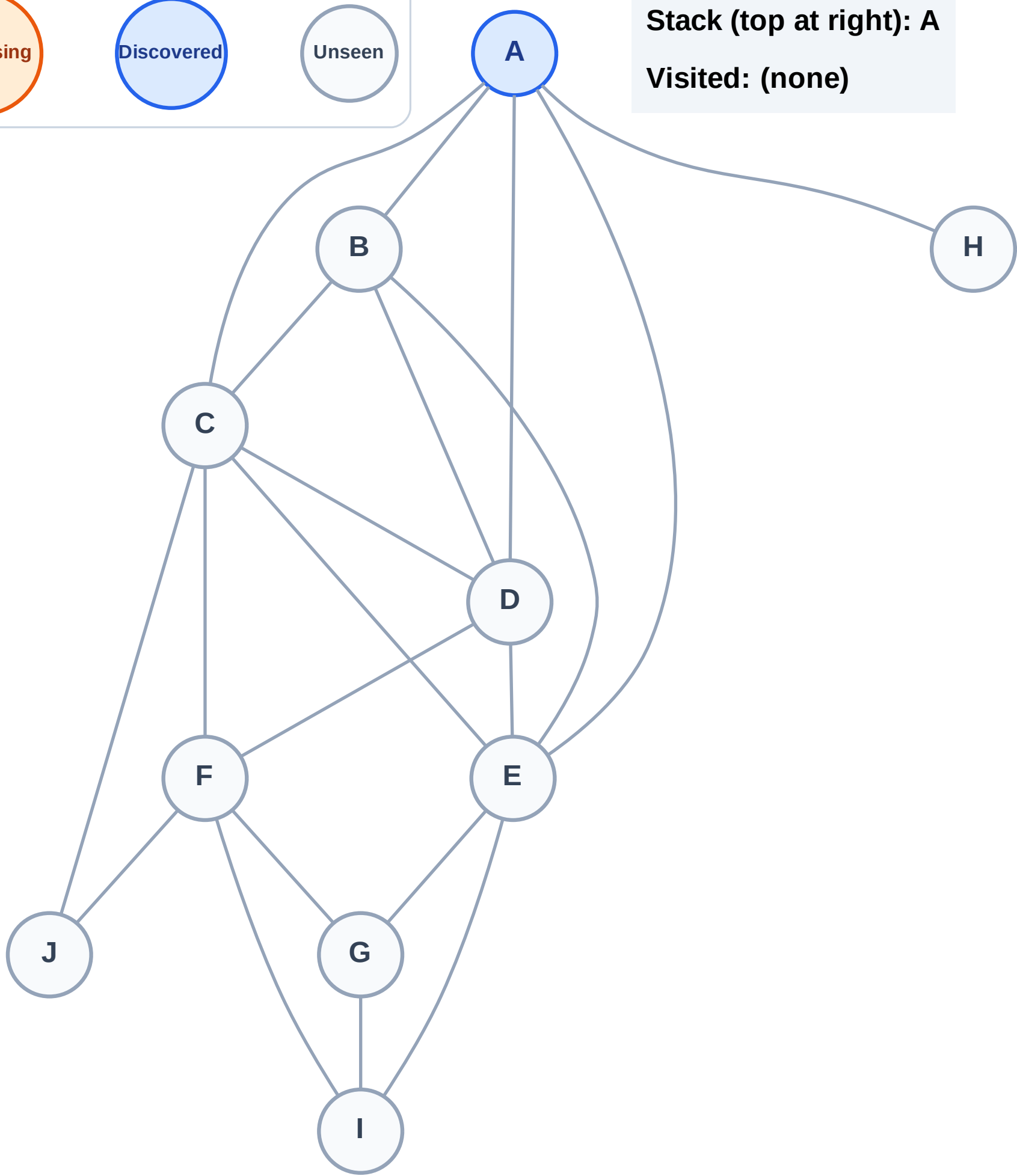
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

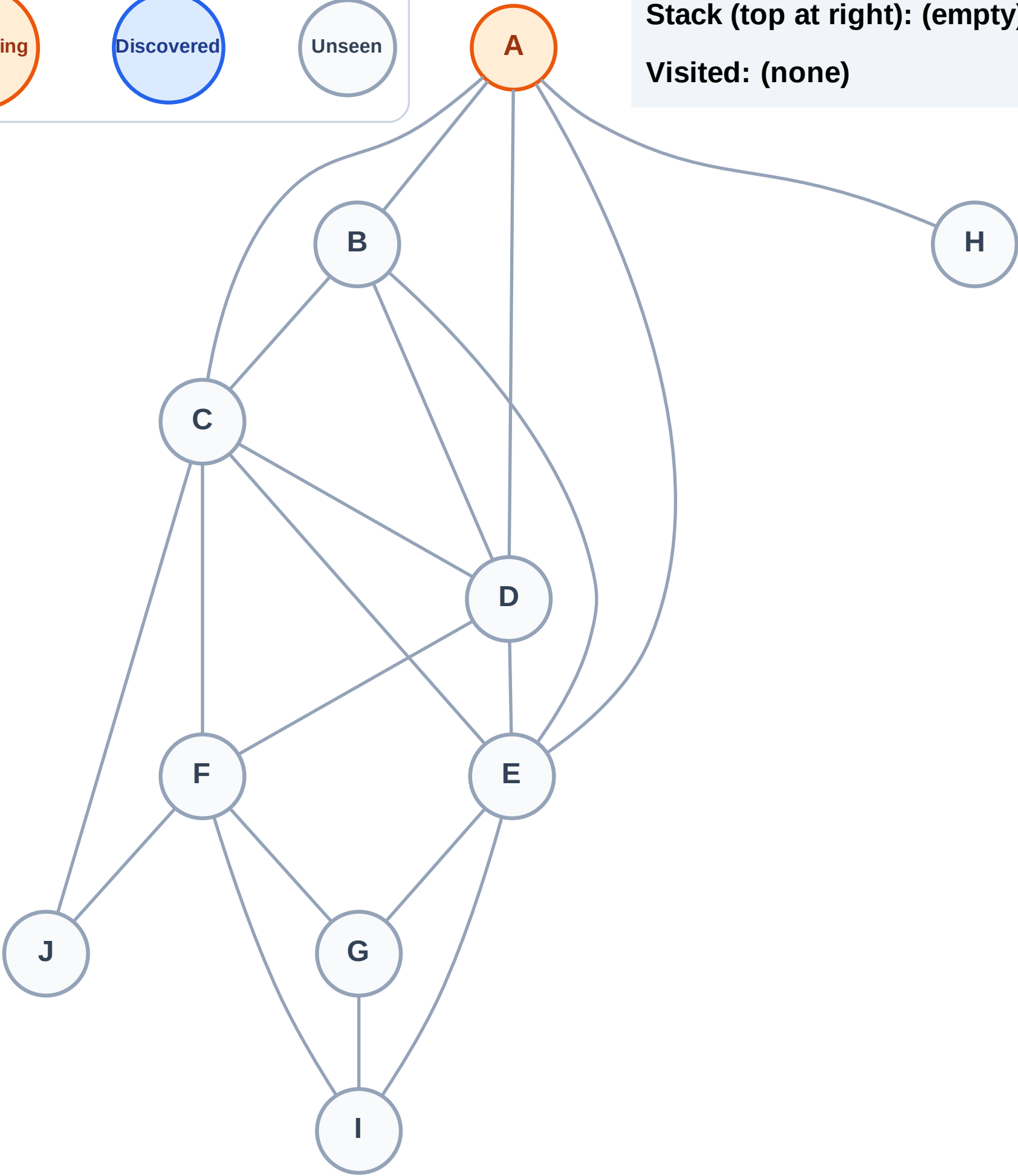
Complete

Processing

Discovered

Unseen

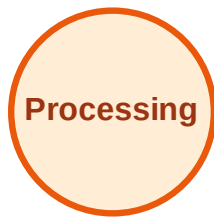
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

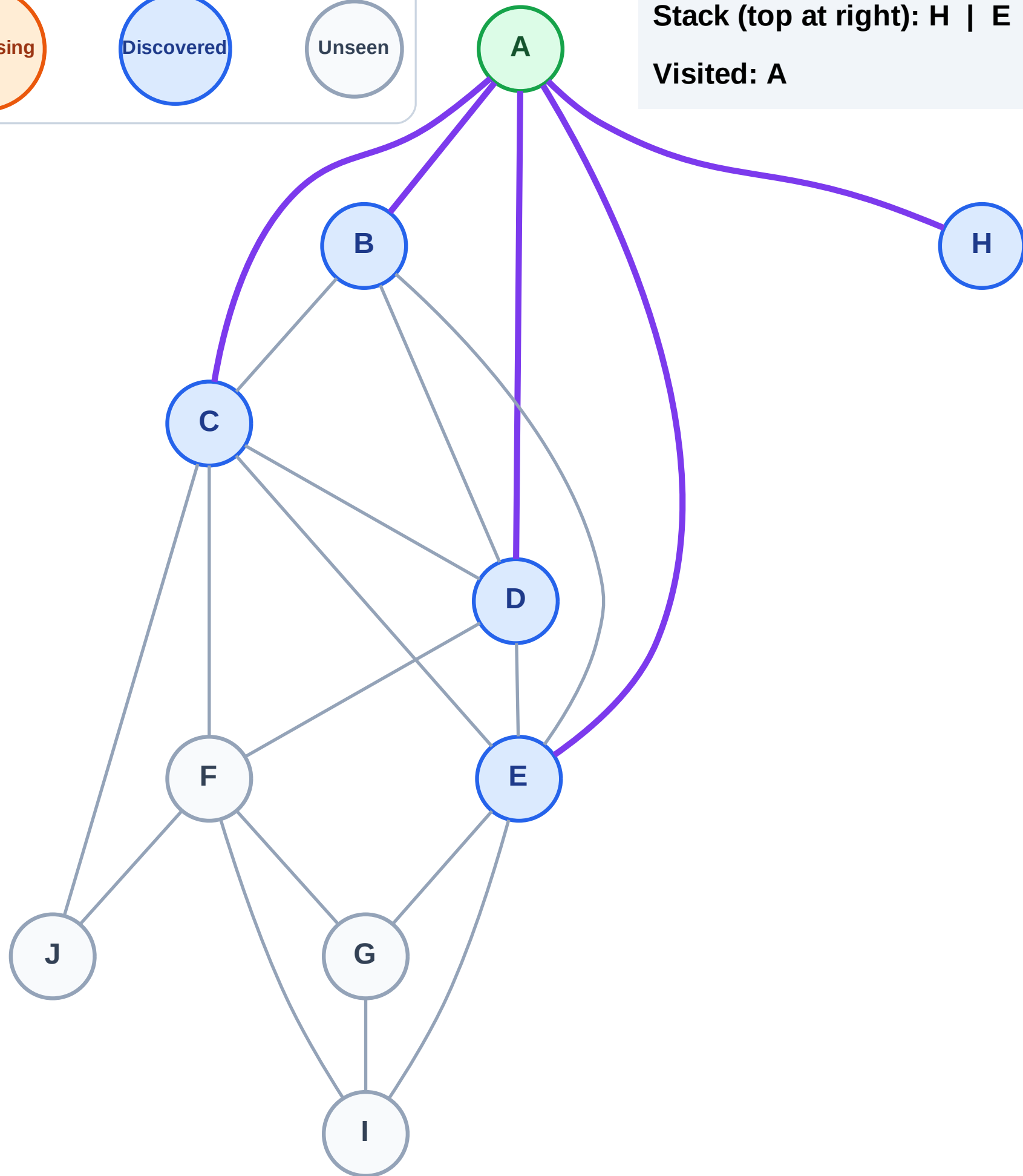
Step 3: Discover H, E, D, C, B from A

Legend



Stack (top at right): H | E | D | C | B

Visited: A



DFS Traversal

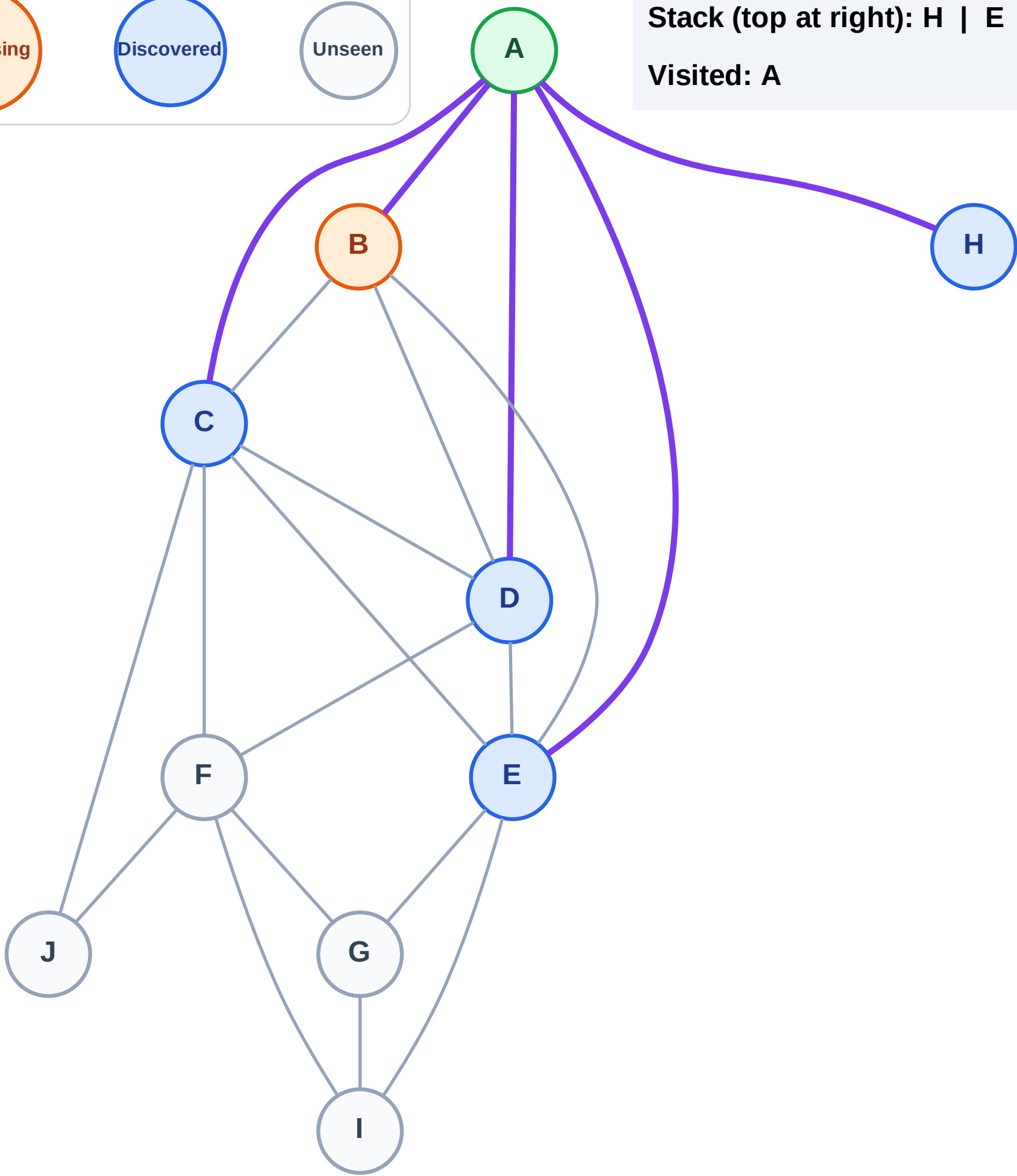
Step 4: Process node B

Legend



Stack (top at right): H | E | D | C

Visited: A



DFS Traversal

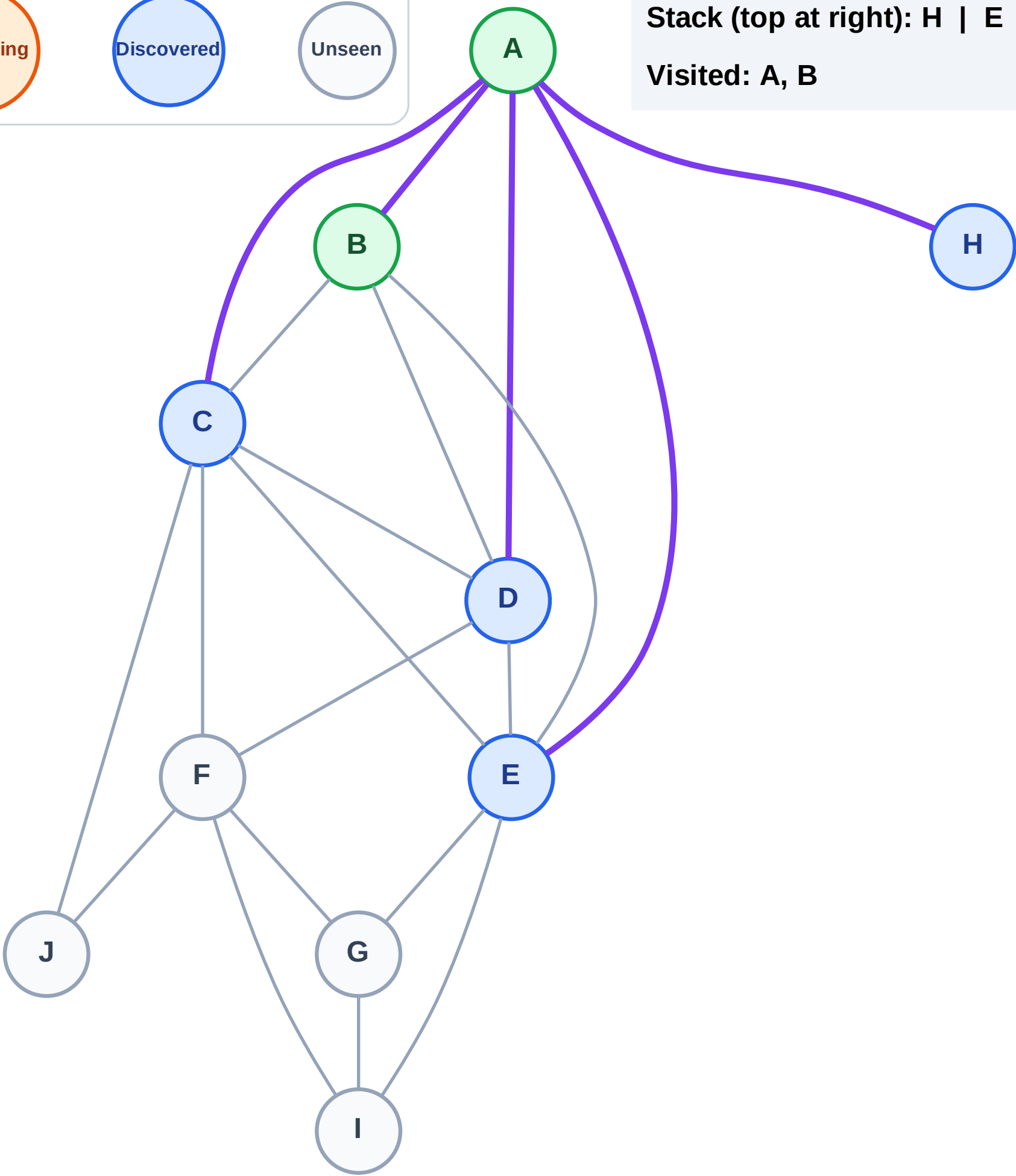
Step 5: No new neighbors from B

Legend



Stack (top at right): H | E | D | C

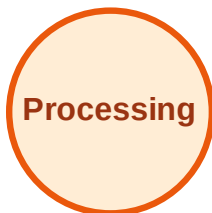
Visited: A, B



DFS Traversal

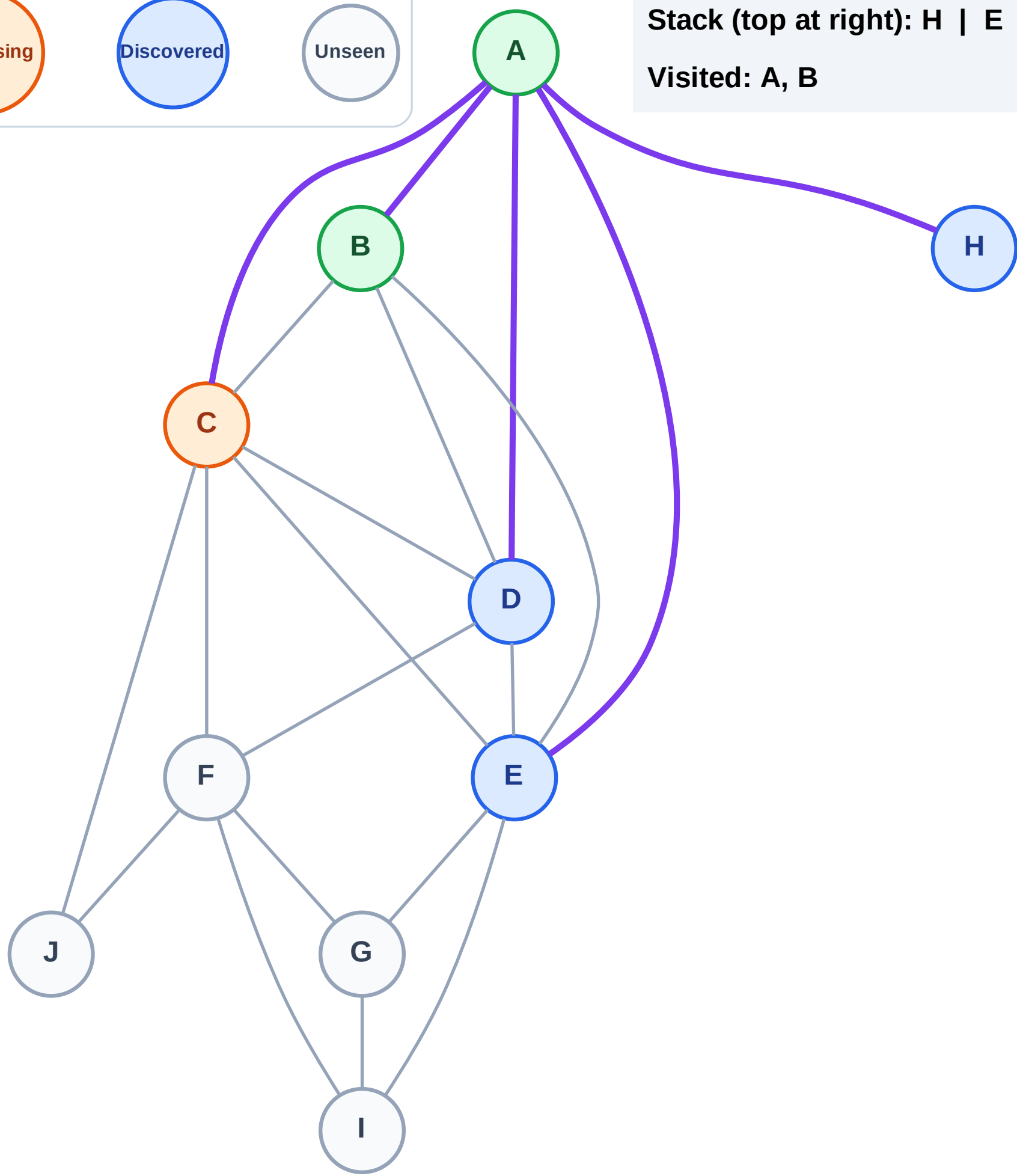
Step 6: Process node C

Legend



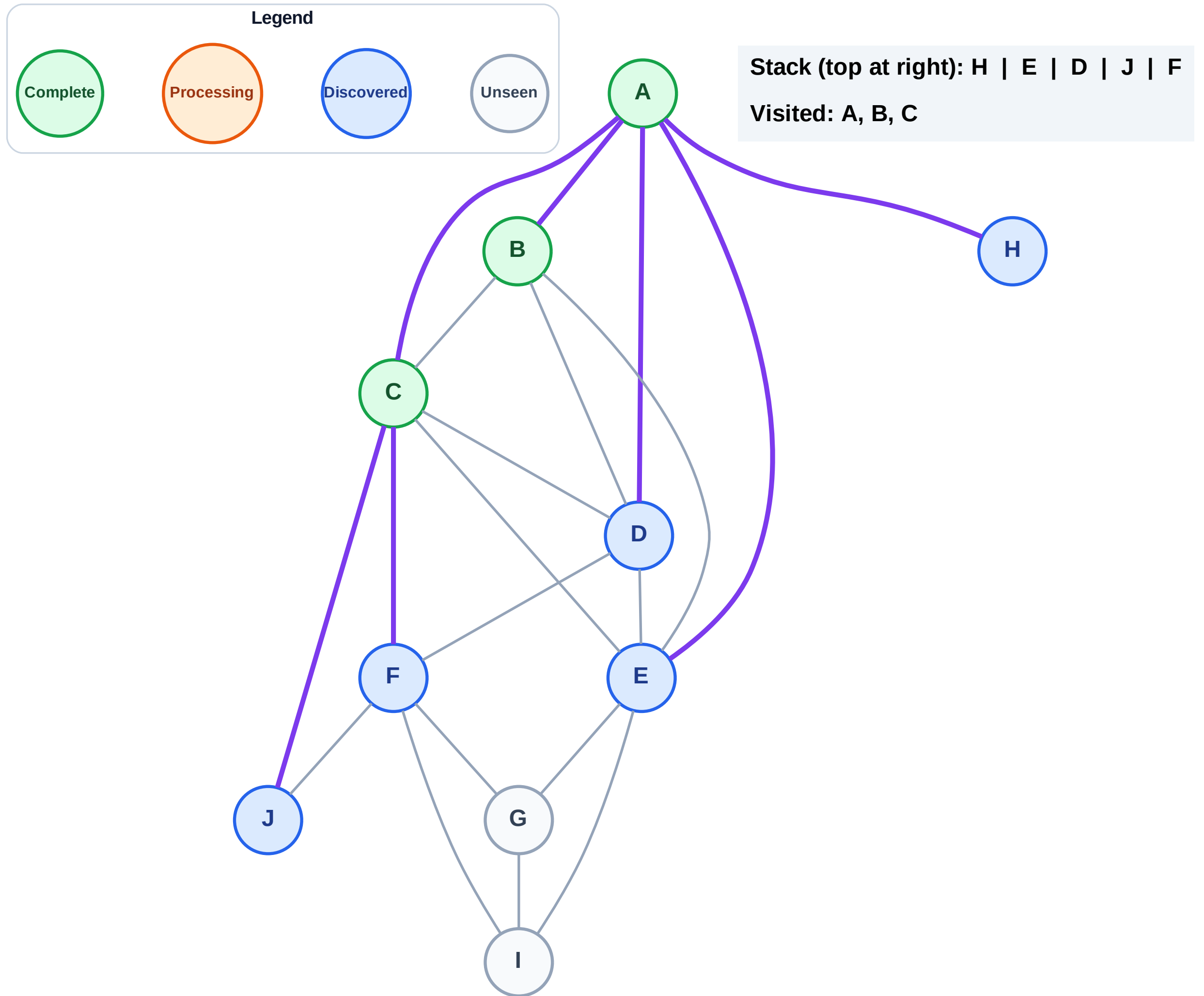
Stack (top at right): H | E | D

Visited: A, B



Step 7: Discover J, F from C

Step 7: Discover J, F from C



DFS Traversal

Step 8: Process node F

Legend

Complete

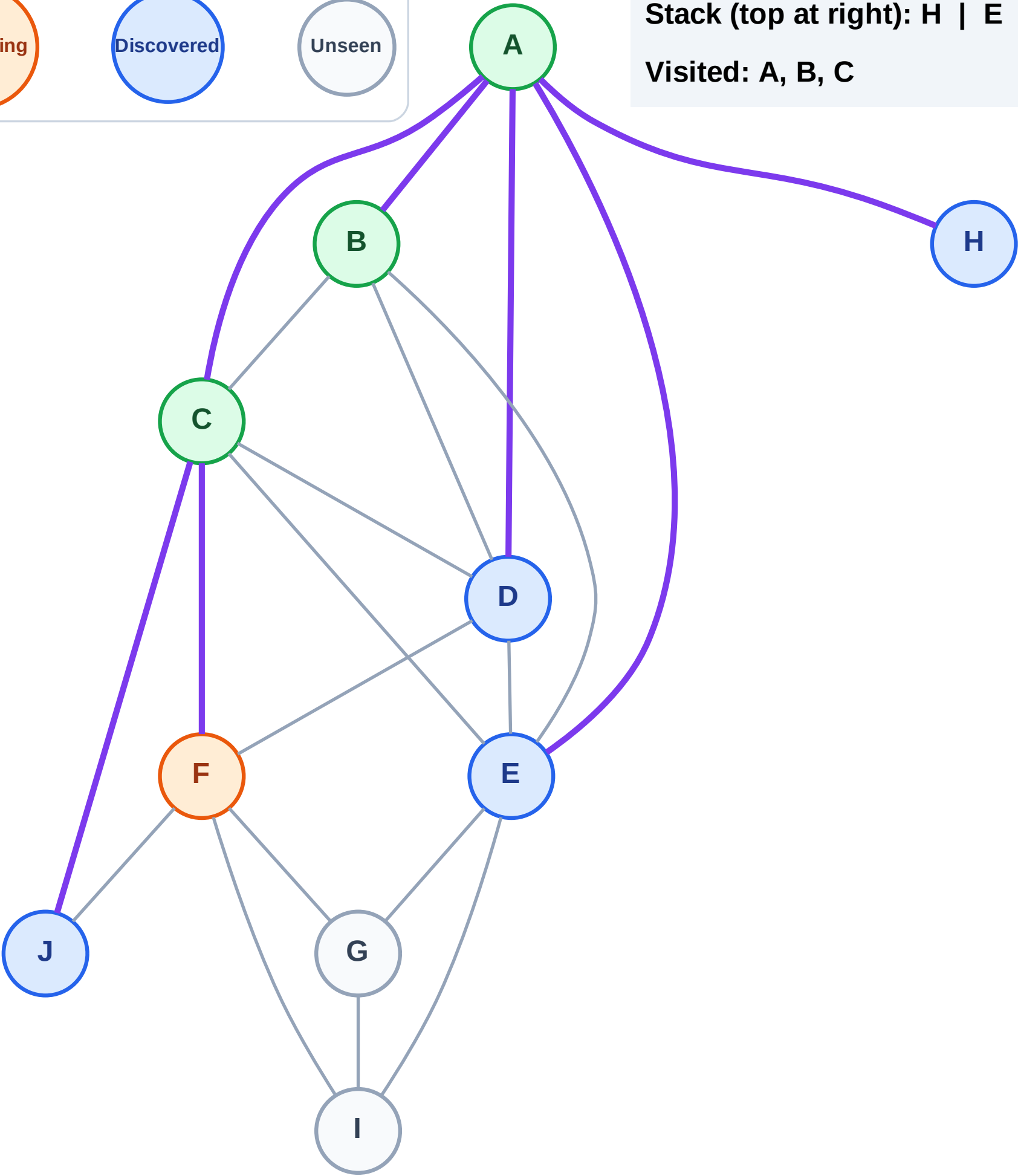
Processing

Discovered

Unseen

Stack (top at right): H | E | D | J

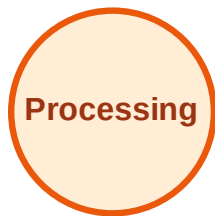
Visited: A, B, C



DFS Traversal

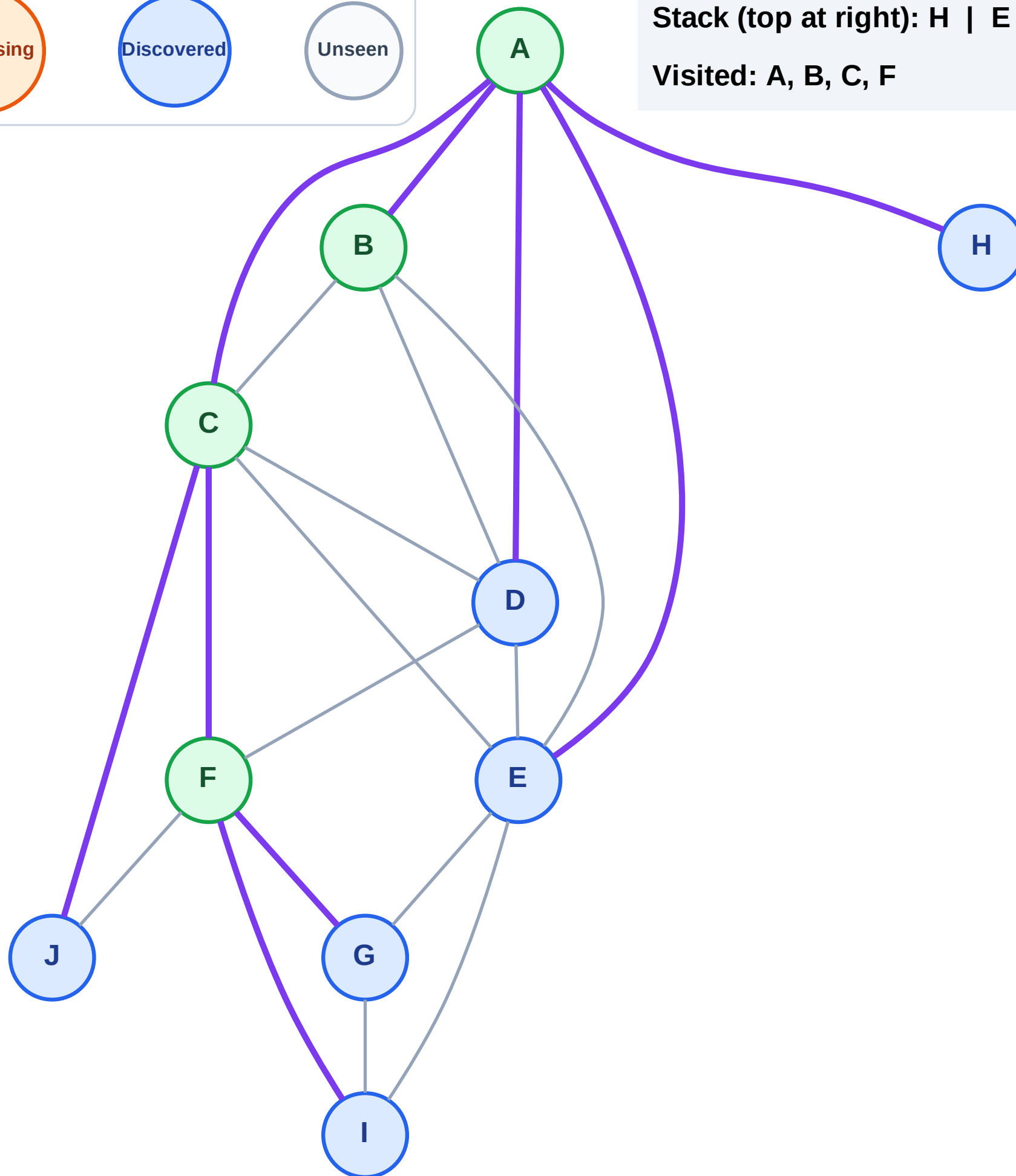
Step 9: Discover I, G from F

Legend



Stack (top at right): H | E | D | J | I | G

Visited: A, B, C, F



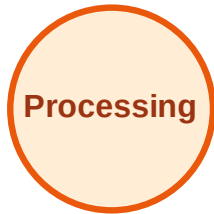
DFS Traversal

Step 10: Process node G

Legend



Complete



Processing



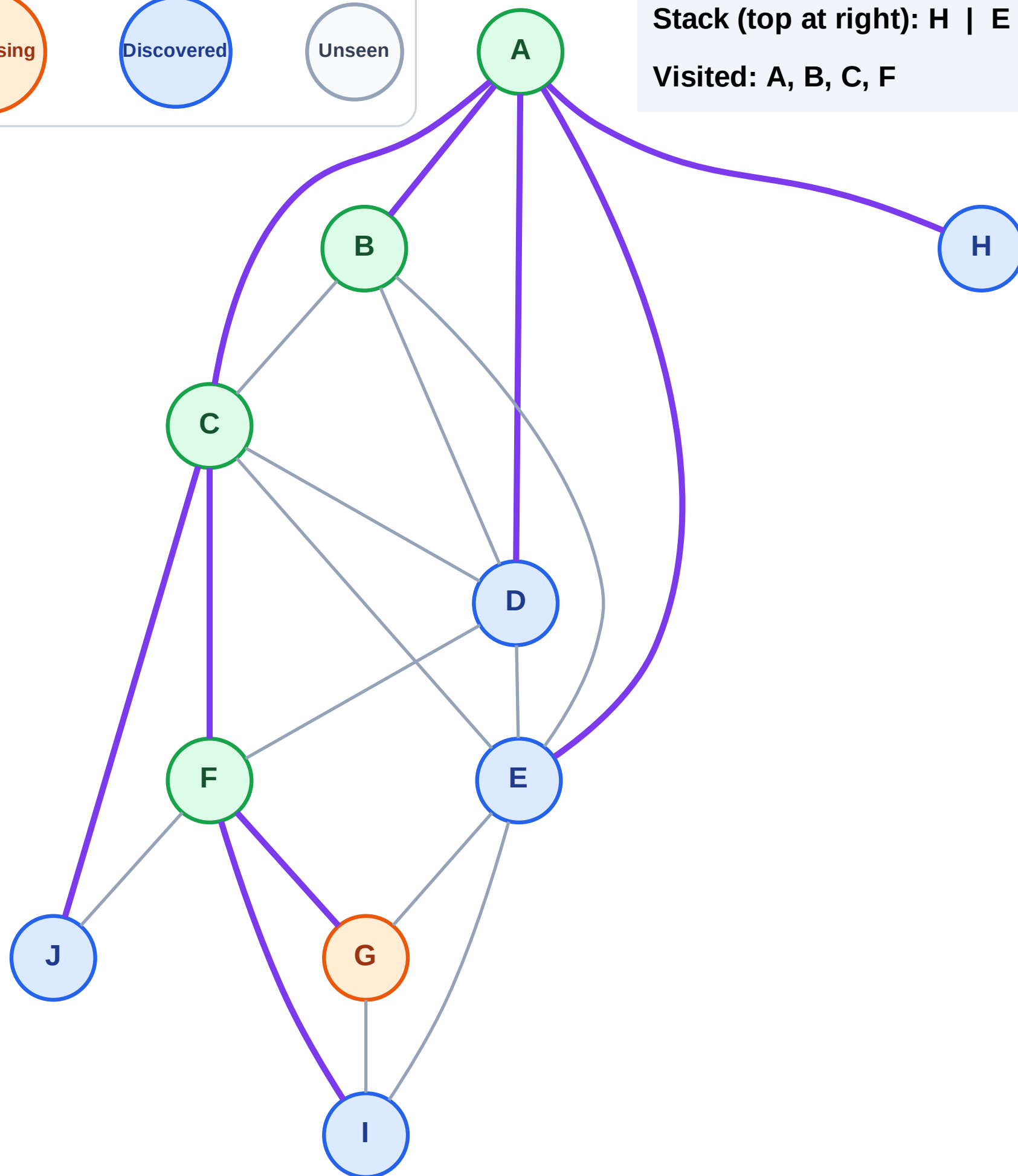
Discovered



Unseen

Stack (top at right): H | E | D | J | I

Visited: A, B, C, F



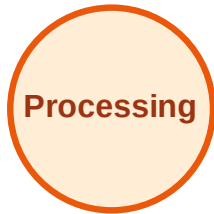
DFS Traversal

Step 11: No new neighbors from G

Legend



Complete



Processing



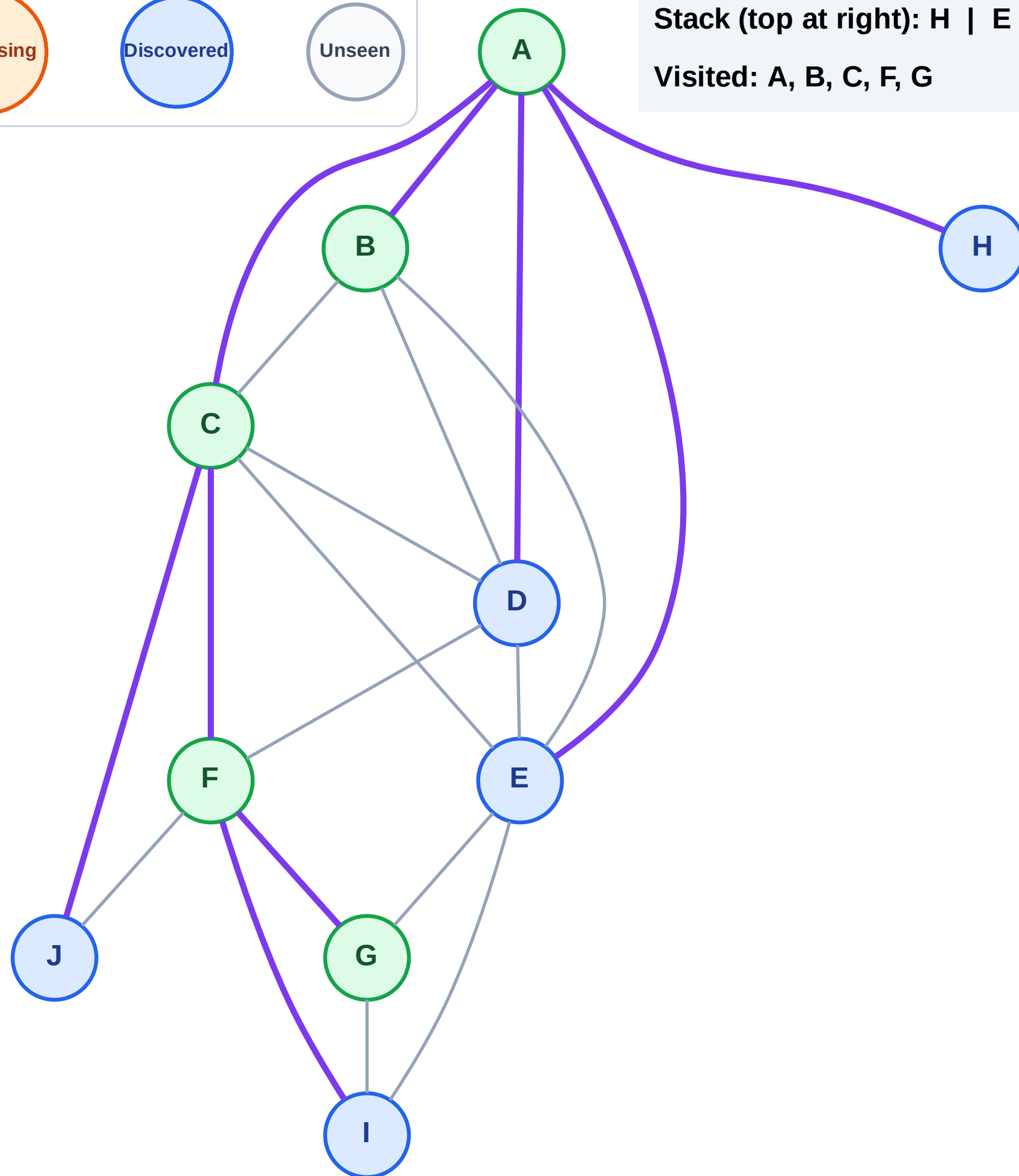
Discovered



Unseen

Stack (top at right): H | E | D | J | I

Visited: A, B, C, F, G



DFS Traversal

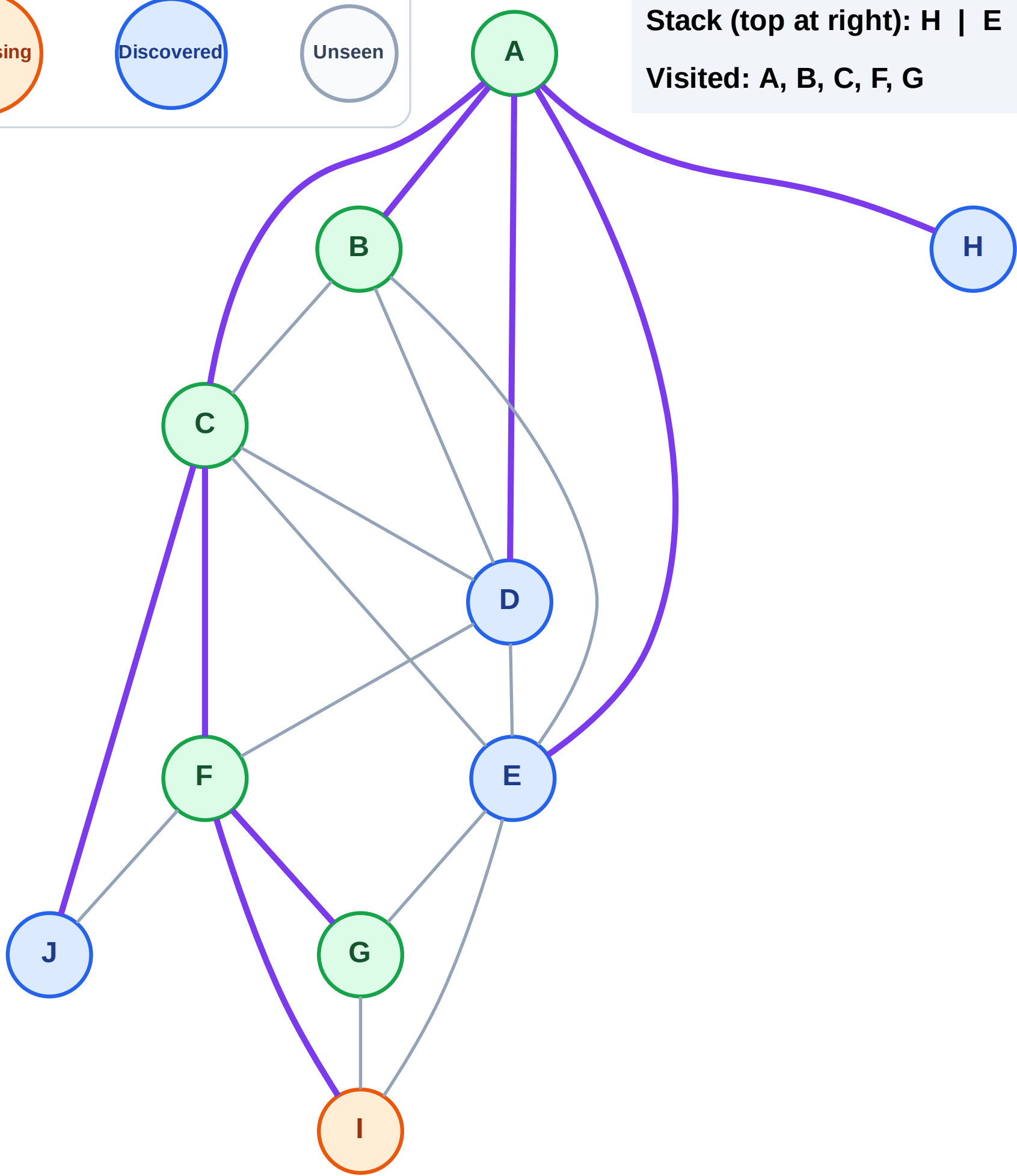
Step 12: Process node I

Legend



Stack (top at right): H | E | D | J

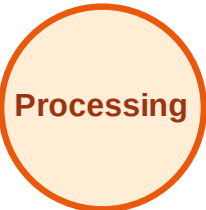
Visited: A, B, C, F, G



DFS Traversal

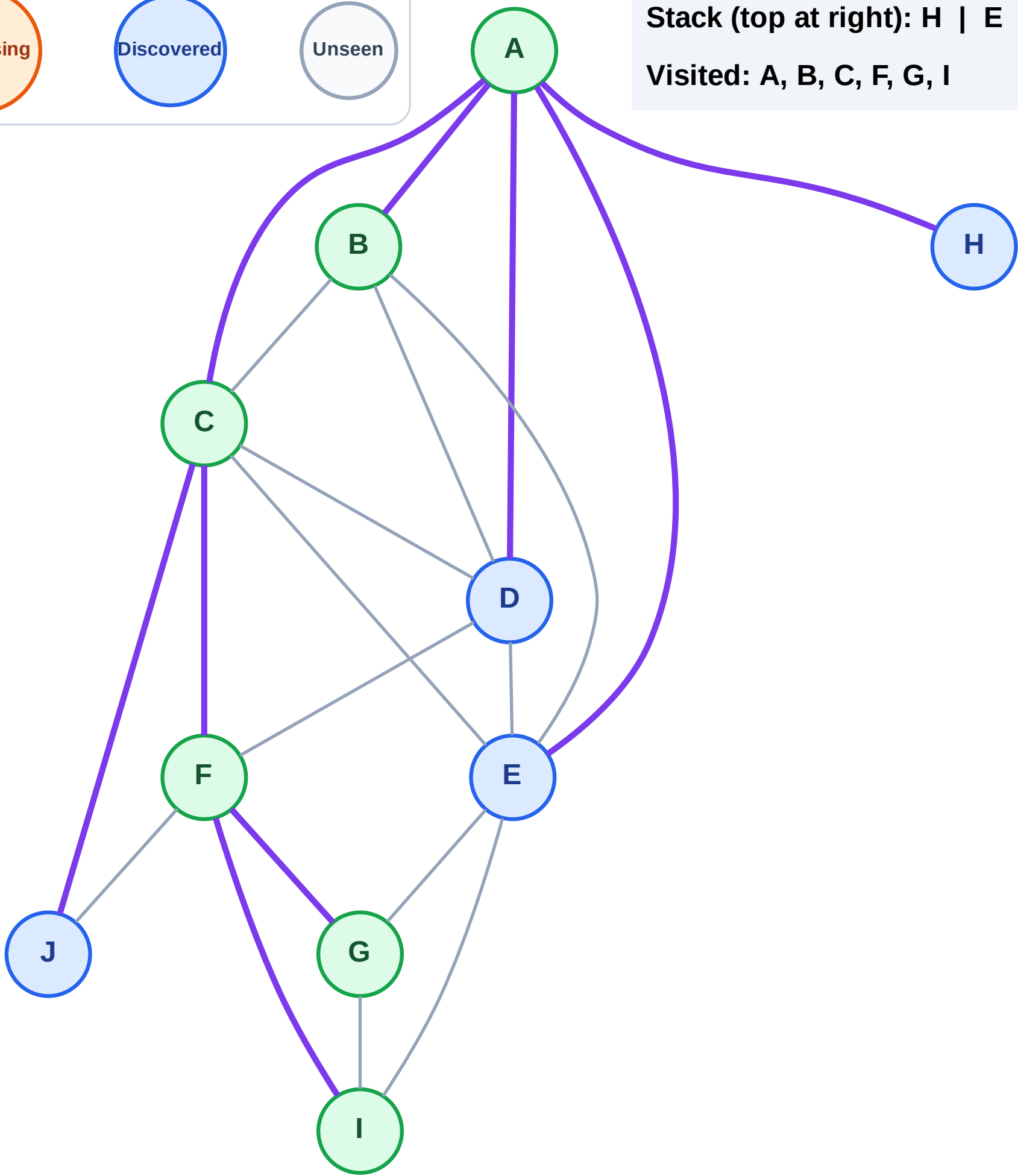
Step 13: No new neighbors from I

Legend



Stack (top at right): H | E | D | J

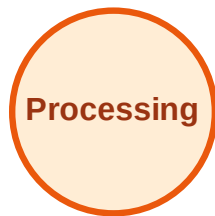
Visited: A, B, C, F, G, I



DFS Traversal

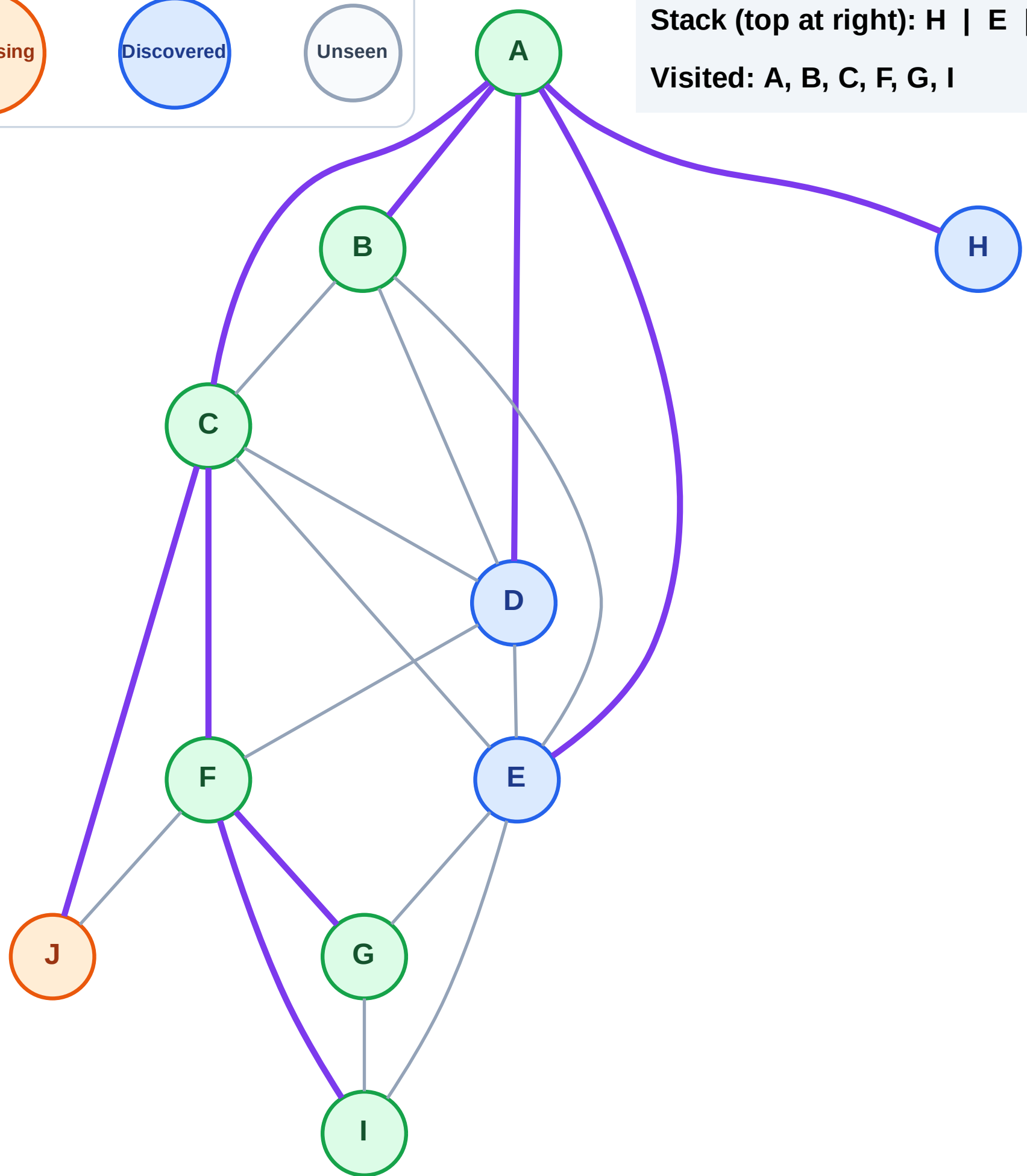
Step 14: Process node J

Legend



Stack (top at right): H | E | D

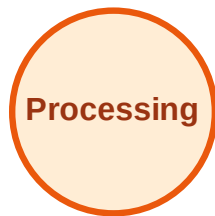
Visited: A, B, C, F, G, I



DFS Traversal

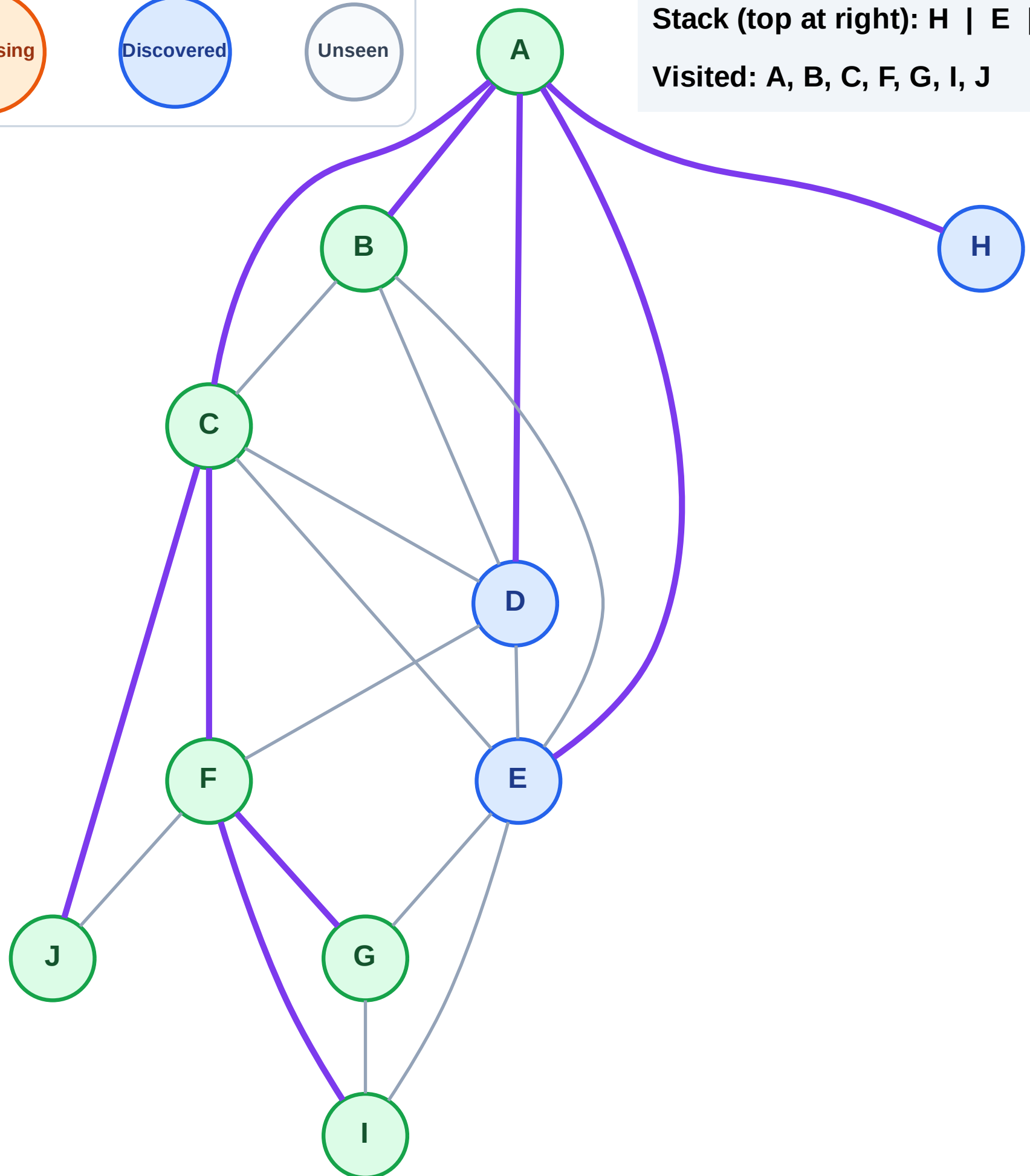
Step 15: No new neighbors from J

Legend



Stack (top at right): H | E | D

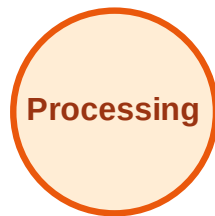
Visited: A, B, C, F, G, I, J



DFS Traversal

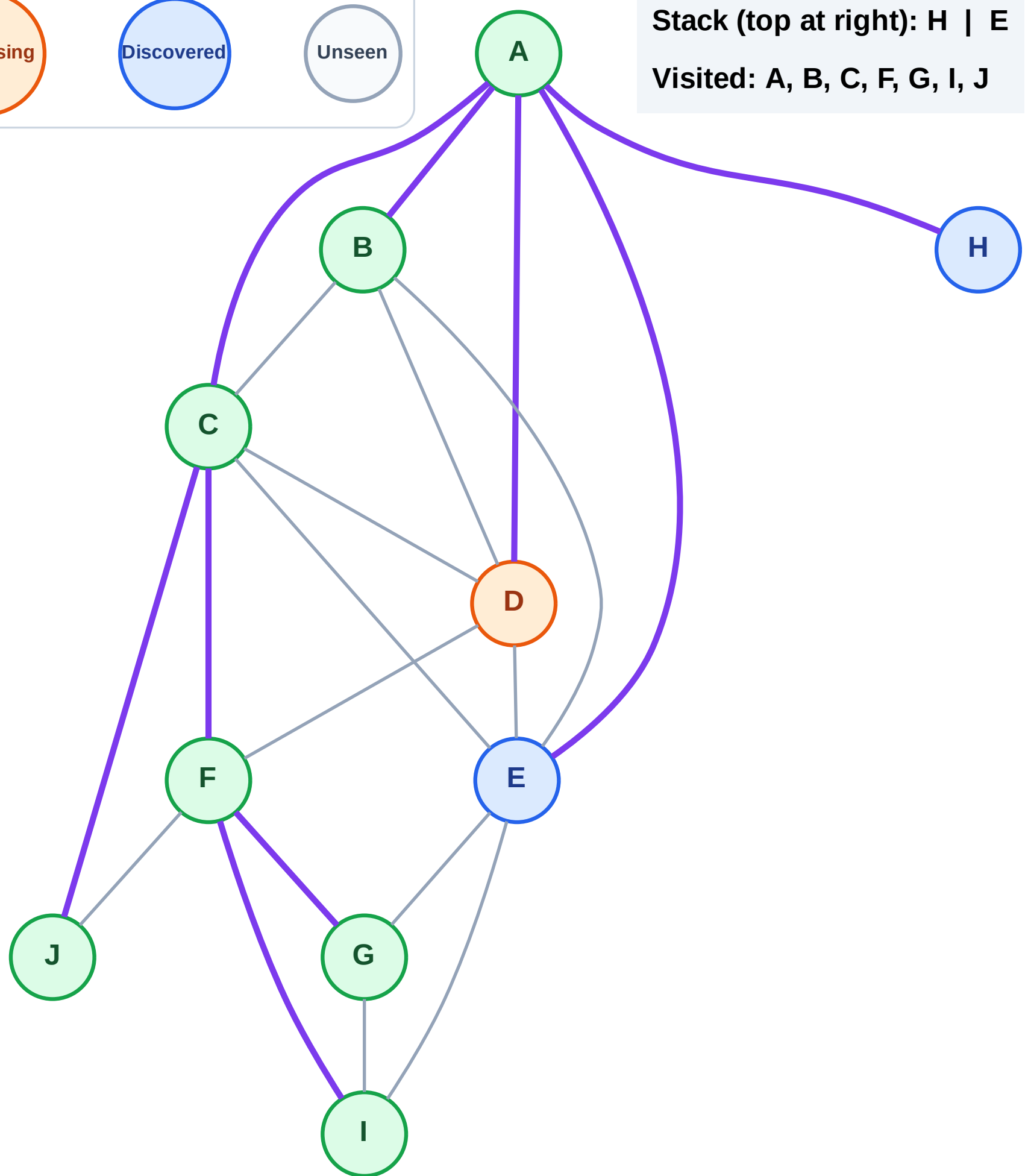
Step 16: Process node D

Legend



Stack (top at right): H | E

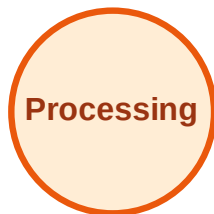
Visited: A, B, C, F, G, I, J



DFS Traversal

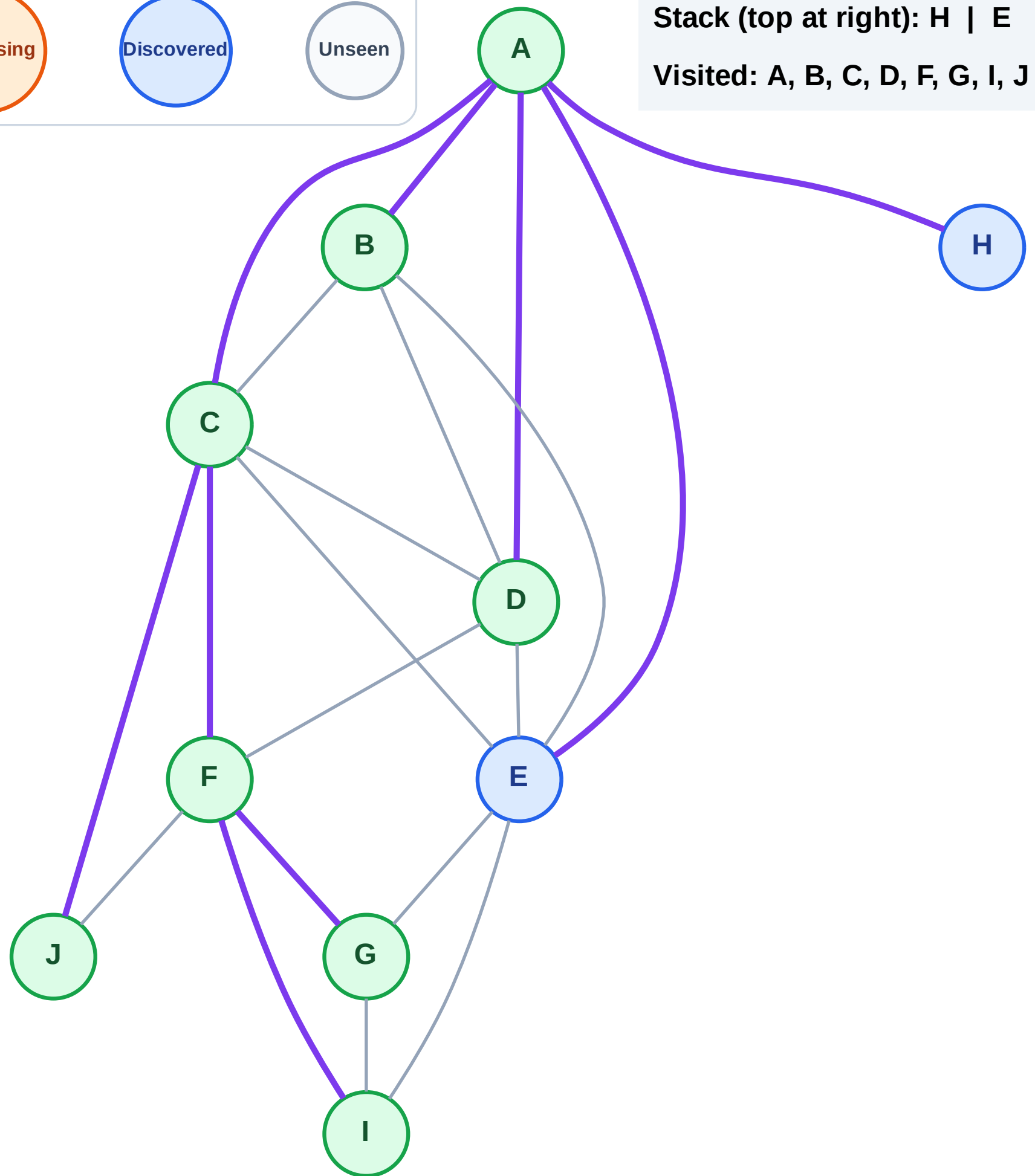
Step 17: No new neighbors from D

Legend



Stack (top at right): H | E

Visited: A, B, C, D, F, G, I, J



DFS Traversal

Step 18: Process node E

Legend

Complete

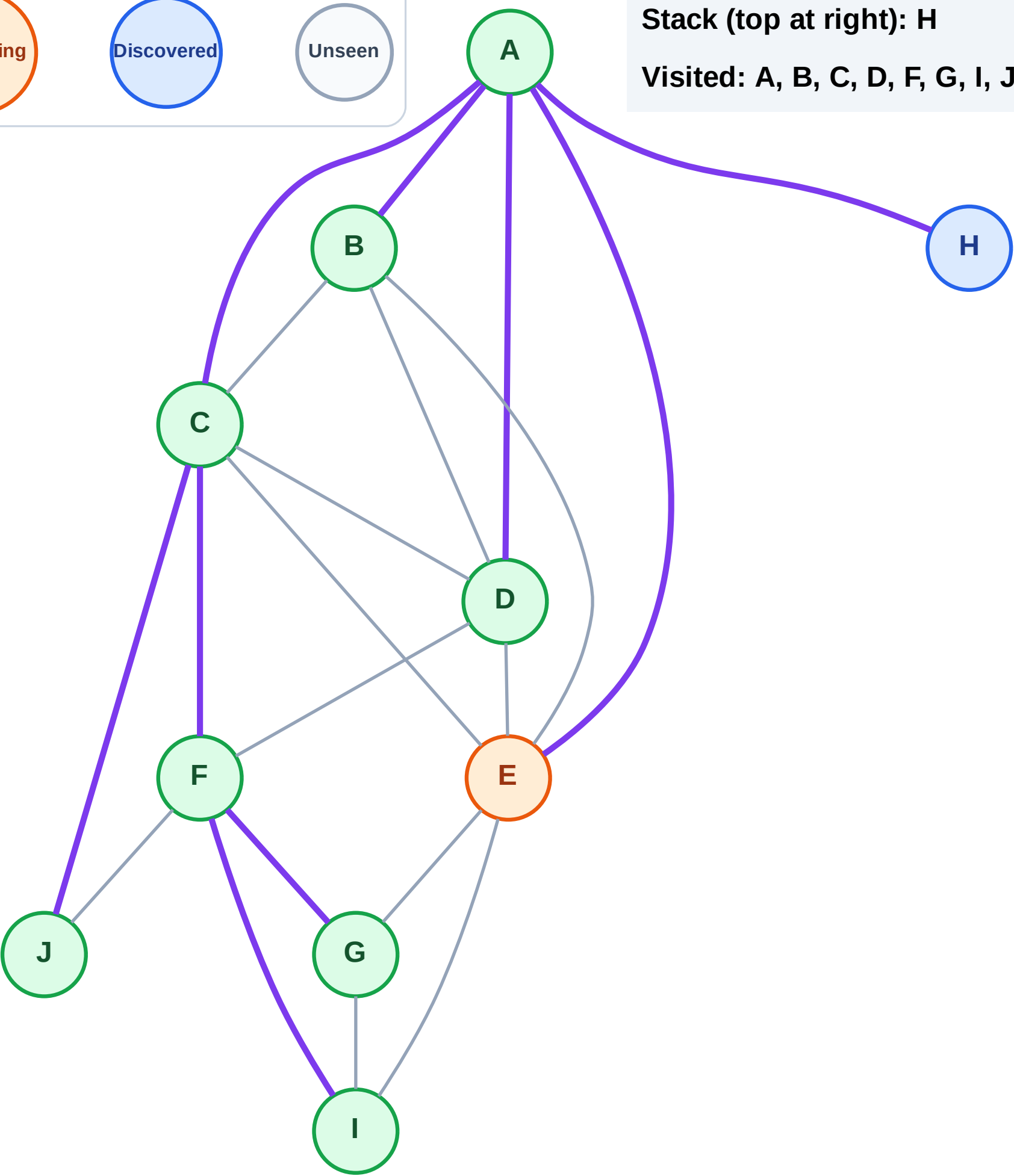
Processing

Discovered

Unseen

Stack (top at right): H

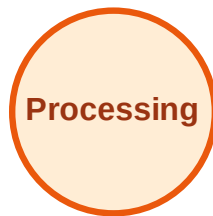
Visited: A, B, C, D, F, G, I, J



DFS Traversal

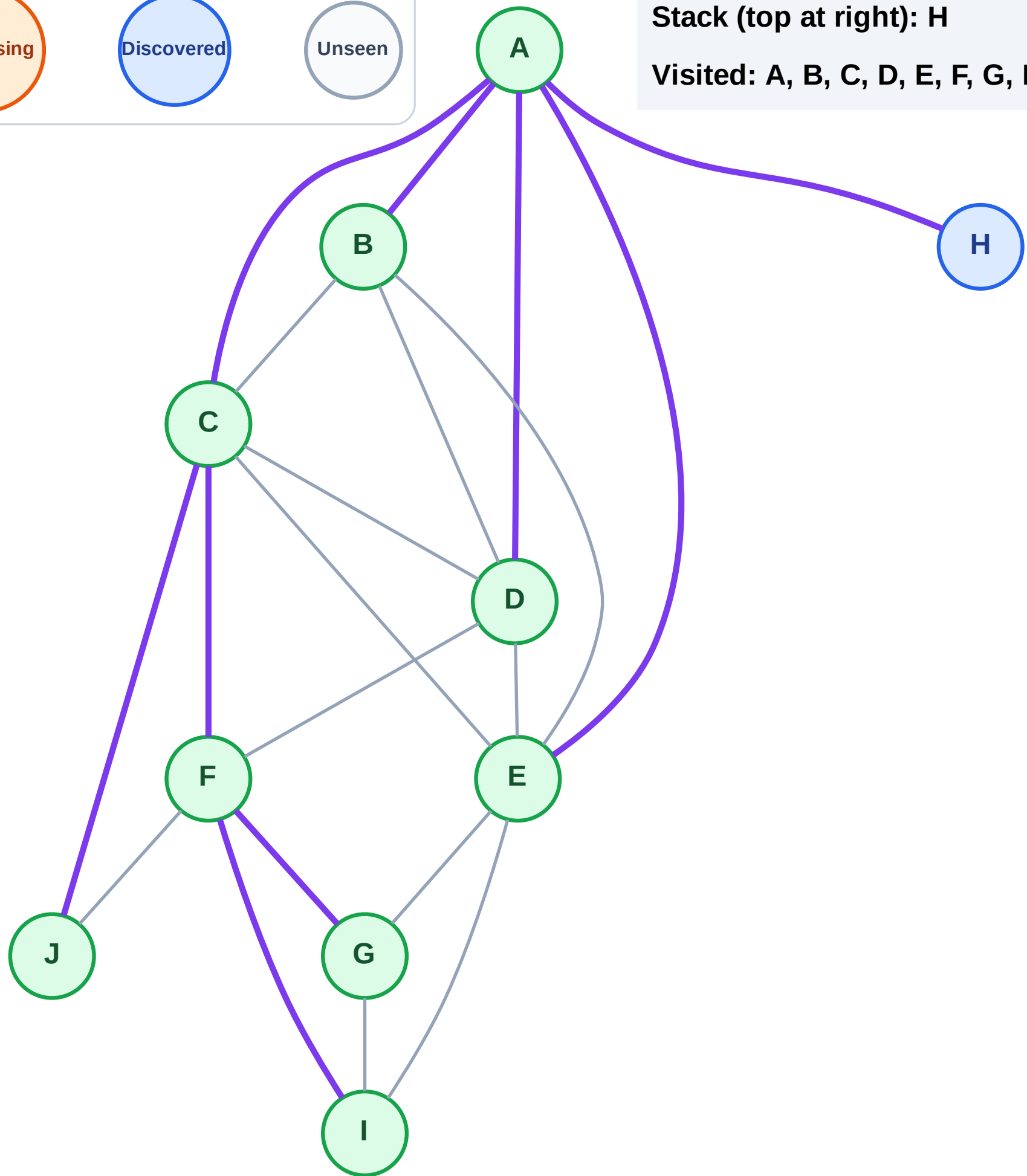
Step 19: No new neighbors from E

Legend



Stack (top at right): H

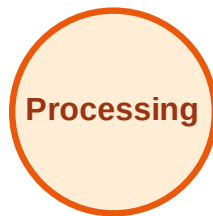
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

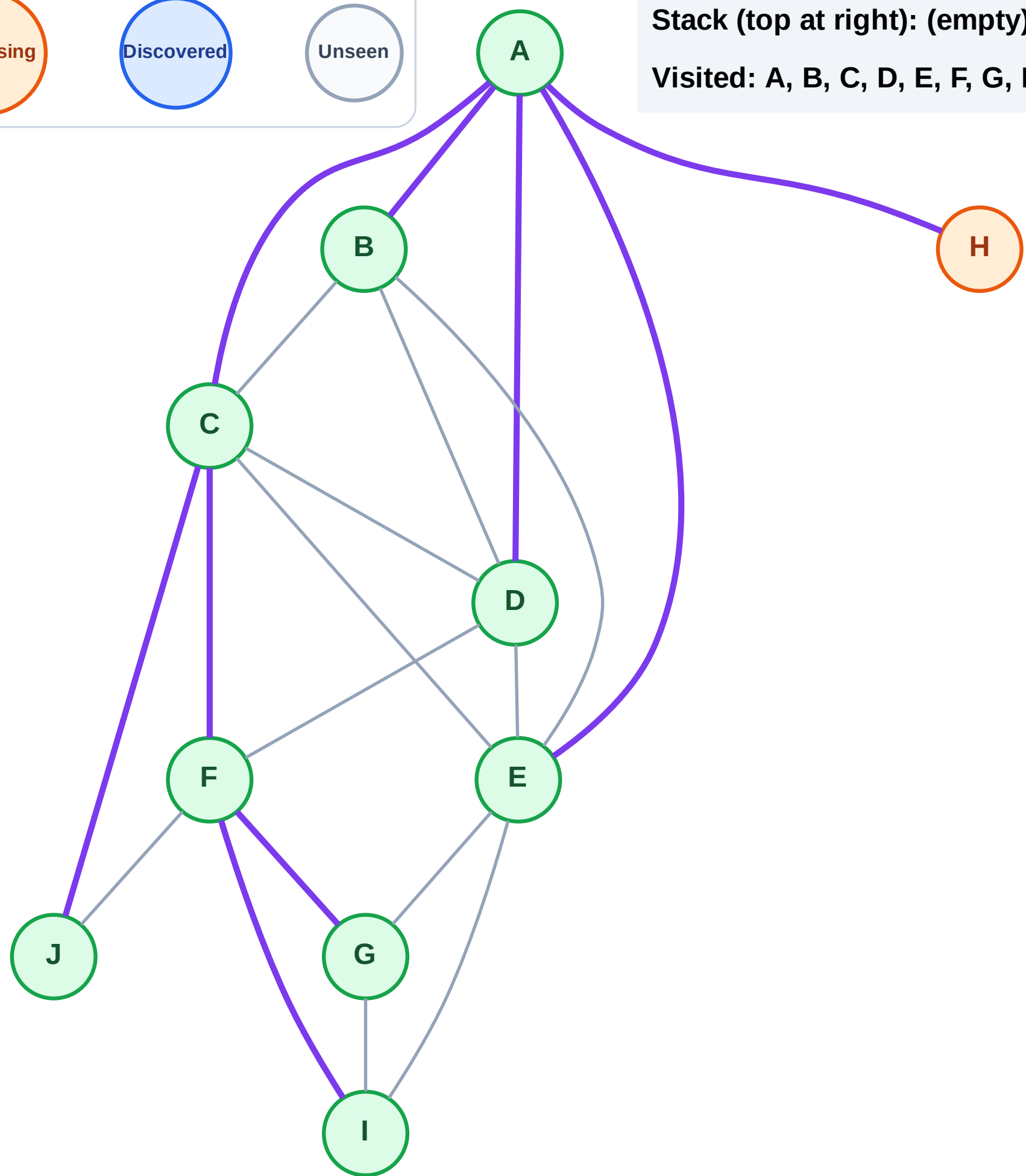
Step 20: Process node H

Legend



Stack (top at right): (empty)

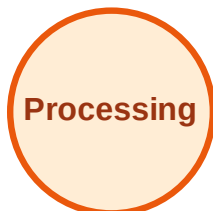
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 21: No new neighbors from H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

